Supplementary materials contains

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Supplemental Table 1. Characteristics of eligible studies for meta-analysis of PD-L1 conversion rates among primary and metastatic sites.

Abbreviation: PD-L1, programmed death ligand 1; TMA, tissue microarray; TPS, tumor proportion score; CPS, combined positive score; IPS, immune cell proportion score; NA, not available.

Study	Year Inclusion	Median age (range)	Specimen sources	IHC antibody (concentration)	Scoring methods	Positivity threshold	Number of observers
Madore 2014	NA	61	Whole tissue	22C3 (1:500)	TPS	1%	3
Callea 2015	NA	58 (40-85)	Whole tissue	405.9A11	TPS	1%	3
Kim 2015	2004-2012	66 (37-88)	TMA	E1L3N	TPS	10%	1
Cimino-Mathews 2016	2001-2011	56 (31-88)	TMA	5H1	TPS	5%	2
Heeren 2016	1985-2008	43 (23-81)	Whole tissue	E1L3N	CPS	5%	NA
Inoue 2016	1990-2013	68 (23-88)	TMA	E1L3N	CPS	5%	NA
Mansfield 2016	1994-2015	61 (54-69)	Whole tissue	E1L3N	IPS	5%	1
Pinato 2016	1970-2005	70 (43-93)	TMA	E1L3N	CPS	5%	2
Straub 2016	2008-2010	58 (38-86)	TMA	E1L3N	TPS	5%	3
Uruga 2016	2002-2010	66 (42-83)	Whole tissue	E1L3N	TPS	1%/5%	3
Kim H 2017	2005-2012	63 (33-77)	Whole tissue	SP142 (1:100)	CPS	5%	2
Kim S 2017	2002-2015	63 (27-84)	Whole tissue	E1L3N	TPS	1%/5%	2
Pichler 2017	NA	69 (36-84)	NA	CLA-10	TPS/IPS	1%	NA
Roper 2017	1997-2015	70 (34-100)	Whole tissue	SP263	TPS	5%	NA
Takamori 2017	2005-2016	63 (29-79)	Whole tissue	SP142 (1:100)	TPS	5%	2
Jong 2018	1998-2017	66	Whole tissue	SP263	IPS	5%	1
Keller 2018	2000-2013	69 (36-85)	TMA	SP142 (1:400)	TPS	50%	1
Miyamoto 2018	1997-2016	63 (41-80)	TMA	SP142 (1:25)	TPS	5%	2
Okada 2018	2000-2016	65 (32-80)	Whole tissue	SP263	TPS	1%	NA
Roussille 2018	2001-2016	64 (35-85)	TMA	NA	TPS	1%	NA
Schneider 2018	2002-2012	58 (27-80)	TMA	5H1 (1:100)	TPS	5%	2
Scognamiglio 2018	NA	NA	Whole tissue	SP263	TPS/IPS	5%	NA
Szekely 2018	NA	NA	Whole tissue/ TMA	E1L3N	CPS	1%	NA
Takamori 2018	2005-2017	64 (42-83)	Whole tissue	SP142	TPS/IPS	1%	2
Tawfik 2018	2007-2016	NA	Whole tissue	SP263	TPS/IPS	1%	NA
Tretiakova 2018	2004-2012	68	TMA	SP142	TPS	1%	2
Yang 2018	NA	69 (54-78)	Whole tissue	SP263	TPS	5%	2
Zhou 2018	2006-2014	57 (26-78)	Whole tissue	6E8	TPS/IPS	5%	2
Alves 2019	1978-1922	59	Whole tissue	SP142	TPS	1%	2
Basu 2019	1995-2016	NA	Whole tissue	22C3	CPS	1%	1
Batur 2019	NA	NA	Whole tissue	22C3	TPS/IPS	1%	1

Erol 2019	2001-2008	53	TMA	E1L3N (1:400)	CPS	5%	2
Manson 2019	NA	53 (27-85)	TMA	SP263	TPS	1%	2
Patel 2019	2006-2017	71 (45-87)	Whole tissue	E1J2J (1:2000)	CPS	1%/10%	NA
Yuan 2019	2017-2018	NA	Whole tissue	NA	CPS	1%	NA
Zhang 2019	2009-2016	54	Whole tissue	NA	TPS	5%	2
Eckel-Passow 2020	1990-2005	62 (35-79)	Whole tissue	NA	CPS	1%	1
Luo 2020	2010-2018	60 (34-80)	Whole tissue	22C3	TPS	1%	NA

Supplemental Table 2. Characteristics of eligible studies for meta-analysis of PD-1 conversion rates among primary and metastatic sites.

Abbreviation: PD-1, programmed death-1; TMA, tissue microarray; TPS, tumor proportion score; CPS, combined positive score; IPS, immune cell proportion score; NA, not available.

Study	Year Inclusion	Median age (range)	Specimen sources	IHC antibody (concentration)	Scoring methods	Positivity threshold	Number of Observers
Miyamoto 2018	1997-2016	63 (41-80)	TMA	NAT105 (1:200)	IPS	1%	2
Roussille 2018	2001-2016	64 (35-85)	TMA	NA	TPS	1%	NA
Schneider 2018	2002-2012	58 (27-80)	TMA	Abcam (1:100)	IPS	5%	2
Basu 2019	1995-2016	NA	Whole tissue	NAT105	IPS	1%	1
Manson 2019	NA	53 (27-85)	TMA	NAT105 (1:200)	IPS	1%	1
Yuan 2019	2017-2018	NA	Whole tissue	NA	IPS	1%	NA
Zhang 2019	2009-2016	54	Whole tissue	ab52587 (1:200)	IPS	5%	2
Eckel-Passow 2020	1990-2005	62 (35-79)	Whole tissue	NA	IPS	1%	1

Supplemental Table 3. Characteristics of eligible studies for meta-analysis of PD-L2 conversion rates among primary and metastatic sites.

Abbreviation: PD-L2, programmed death ligand 2; TMA, tissue microarray; TPS, tumor proportion score; CPS, combined positive score; IPS, immune cell proportion score; NA, not available.

Study	Year Inclusion	Median age (range)	Specimen sources	IHC antibody	Scoring methods	Positivity threshold	Number of Observers
Ogiya 2016	1990-2009	48 (28-64)	Whole tissue	XX19	IPS	10%	1
Pinato 2016	1970-2005	70 (43-93)	TMA	Sigma Aldrich	NA	NA	2
Basu 2019	1995-2016	NA	Whole tissue	3G2	CPS	1%	1
Zhang 2019	2009-2016	54	Whole tissue	ab200377	TPS	5%	2

Supplemental Table 4. Characteristics of eligible studies for meta-analysis of TIL conversion rates among primary and metastatic sites.

Abbreviation: TIL, tumor infiltrating lymphocyte; TMA, tissue microarray; HE, hematoxylin and eosin; NA, not available.

Study	Year Inclusion	Median age (range)	Specimen sources	Scoring methods	Positivity threshold	Number of Observers
Baine 2015	1978-2011	56 (17-72)	TMA	HE staining	5%	2
Mansfield 2016	1994-2015	61 (54-69)	Whole tissue	HE staining	5%	1
Ogiya 2016	1990-2009	48 (28-64)	Whole tissue	HE staining	10%	1
Ogiya 2017	2001-2012	51 (22-73)	Whole tissue	HE staining	10%	1
Eckstein 2018	2001-2017	65 (59-84)	Whole tissue	HE staining	5%	1
Shibutani 2018	2003-2016	67 (33-78)	Whole tissue	HE staining	50%	NA
Szekely 2018	NA	NA	Whole tissue/TMA	HE staining	5%	NA
Zhou 2018	2006-2014	57 (26-78)	Whole tissue	HE staining	50%	2
Batur 2019	NA	NA	Whole tissue	HE staining	NA	1
Kim 2019	2011-2016	59 (32-76)	Whole tissue	HE staining	10%	2
Zhu 2019	NA	49 (39-56)	Whole tissue	Bioinformatics	10%	NA
Hutchinson 2020	2002-2015	51 (34-86)	Whole tissue	Bioinformatics	10%	NA

Supplemental Table 5. Characteristics of eligible studies for tumor mutational burden (TMB) analysis in this study.

Abbreviation: Mb, mega base.

Study	Cancer type	Sample size (pairs)	Metastastic sites	Primary tumor median TMB	Metastatic tumor median TMB	P value
Mansfeld 2018	Lung cancer	13	Brain	12.5 / Mb	24.9 / Mb	<0.001
Tyran 2019	Breast cancer	14	Brain	7.0 / Mb	10.2 / Mb	0.198
He 2020	Synovial sarcoma	7	Lung, liver, pleura	3.3 / Mb	3.2 / Mb	0.362
Hutchinson 2020	Breast cancer	21	Brain, bone, skin	6.0 / Mb	5.0 / Mb	>0.05
Jiang 2020	Lung cancer	20	Brain	10.1 / Mb	10.9 / Mb	>0.05

Supplemental Table 6. Characteristics of eligible studies for microsatellite instability (MSI) analysis in this study.

Abbreviation: PCR, polymerase chain reaction; NGS, next-generation sequencing.

Study	Cancer type	Sample size (pairs)	Metastatic sites	Assessment methods
Messick 2010	Colorectal cancer	21	Lymph node	PCR (BAT25 and BAT26)
Murata 2013	Colorectal cancer	26	Liver, lymph node	PCR (BAT25 and BAT26)
Fujiyoshi 2017	Colorectal cancer	161	Liver	PCR (BAT25, BAT26, D5S346, D2S123 and D17S250)
He 2019	Colorectal cancer	55	Lung, peritoneal, liver	PCR (BAT25, BAT26, D5S346, D2S123 and D17S250)
Sun 2019	Colorectal cancer	33	Brain, liver	NGS (MSIsensor algorithm)
Schlicker 2020	Colorectal cancer	51	Liver	NGS (64-gene signature)

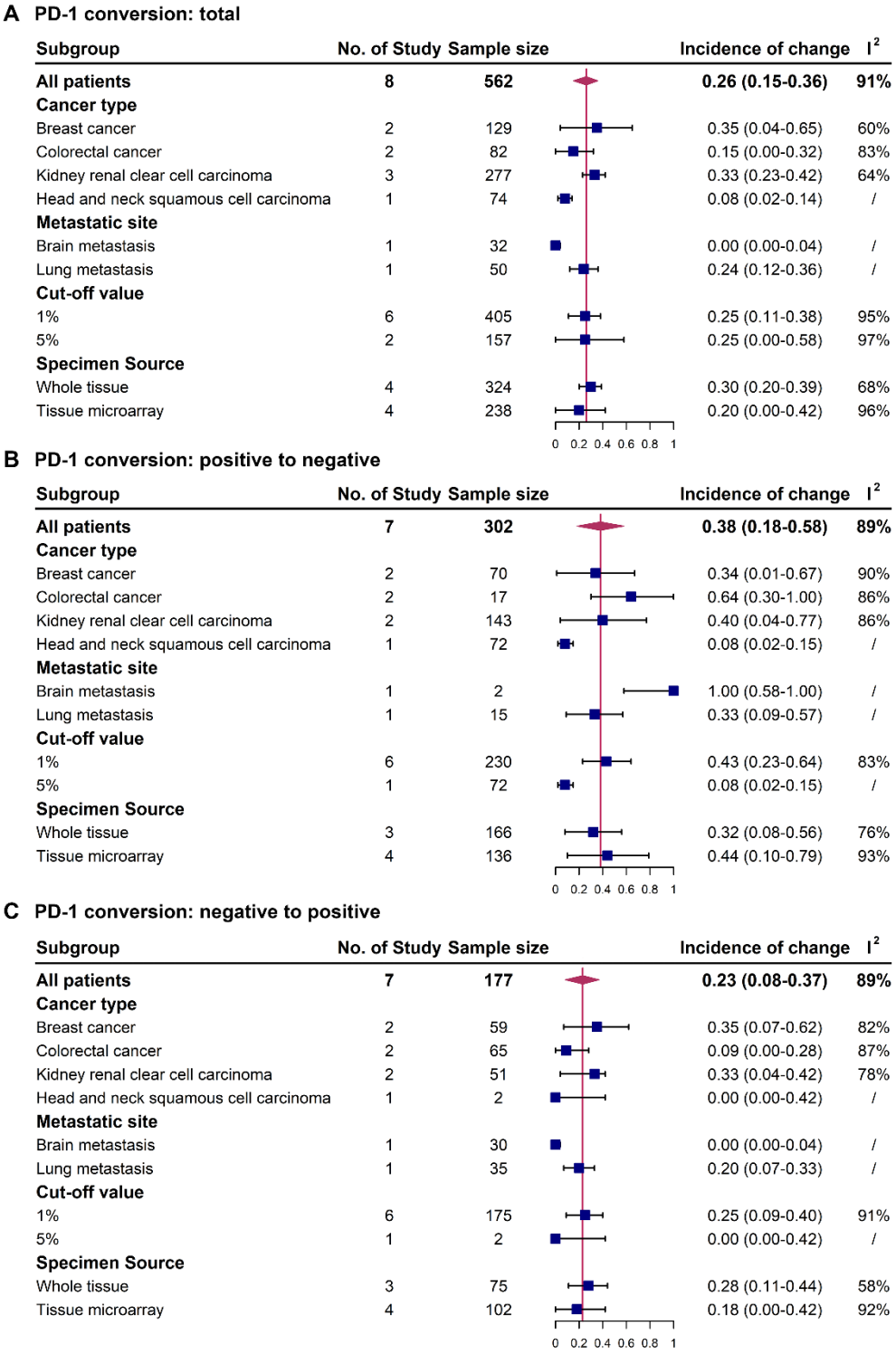
Supplemental Table 7. Methodological quality summary of included studies according to QUADAS-2 tool.

Abbreviations: +, low risk; -, high risk; ?, unclear risk.

Study	Risk of bias				Concerns regarding applicability		
	Patient selection	Index test	Reference standard	Flow and timing	Patient selection	Index test	Reference standard
Messick 2010	?	+	+	+	+	+	+
Murata 2013	+	+	+	+	+	+	+
Madore 2014	?	+	+	?	+	+	+
Baine 2015	+	+	+	+	+	+	+
Callea 2015	?	+	+	+	+	?	+
Kim 2015	+	+	+	+	+	+	+
Cimino-Mathews 2016	+	+	+	+	+	+	+
Heeren 2016	+	+	+	?	+	+	?
Inoue 2016	+	+	+	+	+	+	+
Mansfield 2016	+	+	+	+	+	+	+
Ogiya 2016	+	+	+	+	+	+	+
Pinato 2016	+	-	?	+	+	-	+
Straub 2016	+	+	+	+	+	+	+
Uruga 2016	+	+	+	+	?	+	+
Fujiyoshi 2017	+	+	+	+	+	+	+
Kim H 2017	+	+	+	+	+	+	?
Kim S 2017	+	+	+	+	+	+	+
Ogiya 2017	+	+	+	+	+	+	+
Pichler 2017	-	+	+	+	+	+	+
Roper 2017	+	+	+	-	+	+	+
Takamori 2017	+	+	+	+	+	+	+
Eckstein 2018	+	+	+	?	+	+	+
Jong 2018	+	+	+	+	+	+	+
Keller 2018	+	+	+	?	+	+	-
Mansfield 2018	?	+	+	+	+	+	+
Miyamoto 2018	+	+	+	+	+	+	+
Okada 2018	+	+	+	+	+	+	+
Roussille 2018	+	+	+	+	+	+	+
Schneider 2018	+	+	+	?	+	+	+
Scognamiglio 2018	?	+	+	+	-	+	+
Shibutani 2018	+	+	+	+	+	?	+
Szekely 2018	?	+	+	+	+	+	+
Takamori 2018	+	+	+	+	+	+	+
Tawfik 2018	+	+	+	+	+	+	+
Tretiakova 2018	+	+	+	?	+	+	+
Yang 2018	?	+	+	+	-	+	+
Zhou 2018	+	+	+	+	+	+	+
Alves 2019	+	+	+	+	+	+	+
Basu 2019	+	+	+	+	+	+	+
Batur 2019	?	+	+	+	?	-	+
Erol 2019	+	+	+	+	+	+	+

He 2019	+	+	+	+	+	+	+
Kim 2019	+	+	+	+	+	+	+
Manson 2019	?	+	+	+	+	+	+
Patel 2019	+	+	+	+	+	+	+
Sun 2019	?	+	+	+	+	+	+
Tyran 2019	?	+	+	+	+	+	+
Yuan 2019	+	+	+	?	+	+	+
Zhang 2019	+	+	+	+	+	+	+
Zhu 2019	?	+	+	+	?	+	+
Eckel-Passow 2020	+	+	+	+	+	+	+
Luo 2020	+	+	+	?	+	+	-
He 2020	+	+	+	+	+	+	+
Hutchinson 2020	+	+	+	+	+	+	+
Jiang 2020	+	+	+	+	+	+	+
Schlicker 2020	?	+	+	+	+	+	?

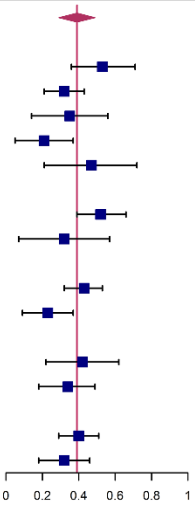
Supplementary Figures



Supplemental Figure S1. Subgroup analysis of PD-1 conversion rates based on IHC positivity threshold, assessment methods, specimen sources, the number of observers, and the time of metastasis. Disaccordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.

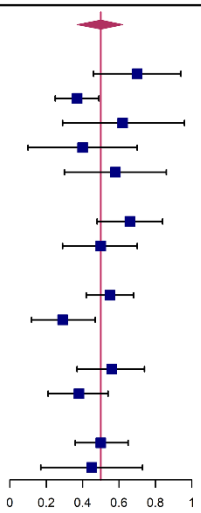
A TIL conversion: total

Subgroup	No. of Study	Sample size	Incidence of change	I^2
All patients	**12**	**333**	**0.39 (0.29-0.49)**	**77%**
Cancer type				
Non-small cell lung cancer	4	94	0.53 (0.36-0.71)	72%
Breast cancer	5	180	0.32 (0.21-0.43)	65%
Kidney renal clear cell carcinoma	1	20	0.35 (0.14-0.56)	/
Colorectal cancer	1	24	0.21 (0.05-0.37)	/
Bladder cancer	1	15	0.47 (0.21-0.72)	/
Metastatic site				
Brain metastasis	5	140	0.52 (0.39-0.66)	66%
Liver metastasis	2	39	0.32 (0.07-0.57)	65%
TIL assessment				
HE staining	10	247	0.43 (0.32-0.53)	70%
Bioinformatics	2	86	0.23 (0.09-0.37)	61%
Cut-off value				
5%	5	115	0.42 (0.22-0.62)	66%
10%	4	121	0.34 (0.18-0.49)	74%
Specimen Source				
Whole tissue	10	290	0.40 (0.29-0.51)	81%
Tissue microarray	2	43	0.32 (0.18-0.46)	0%



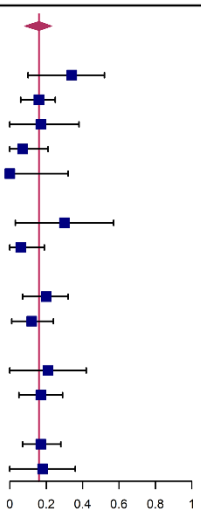
B TIL conversion: high to low

Subgroup	No. of Study	Sample size	Incidence of change	I^2
All patients	**11**	**204**	**0.50 (0.37-0.62)**	**76%**
Cancer type				
Non-small cell lung cancer	3	46	0.70 (0.46-0.94)	66%
Breast cancer	5	128	0.37 (0.25-0.49)	54%
Kidney renal clear cell carcinoma	1	8	0.62 (0.29-0.96)	/
Colorectal cancer	1	10	0.40 (0.10-0.70)	/
Bladder cancer	1	12	0.58 (0.30-0.86)	/
Metastatic site				
Brain metastasis	4	82	0.66 (0.48-0.84)	68%
Liver metastasis	2	22	0.50 (0.29-0.70)	0%
TIL assessment				
HE staining	9	150	0.55 (0.42-0.68)	67%
Bioinformatics	2	54	0.29 (0.12-0.47)	55%
Cut-off value				
5%	5	72	0.56 (0.37-0.74)	69%
10%	4	86	0.38 (0.21-0.54)	65%
Specimen Source				
Whole tissue	9	178	0.50 (0.36-0.65)	69%
Tissue microarray	2	26	0.45 (0.17-0.73)	51%

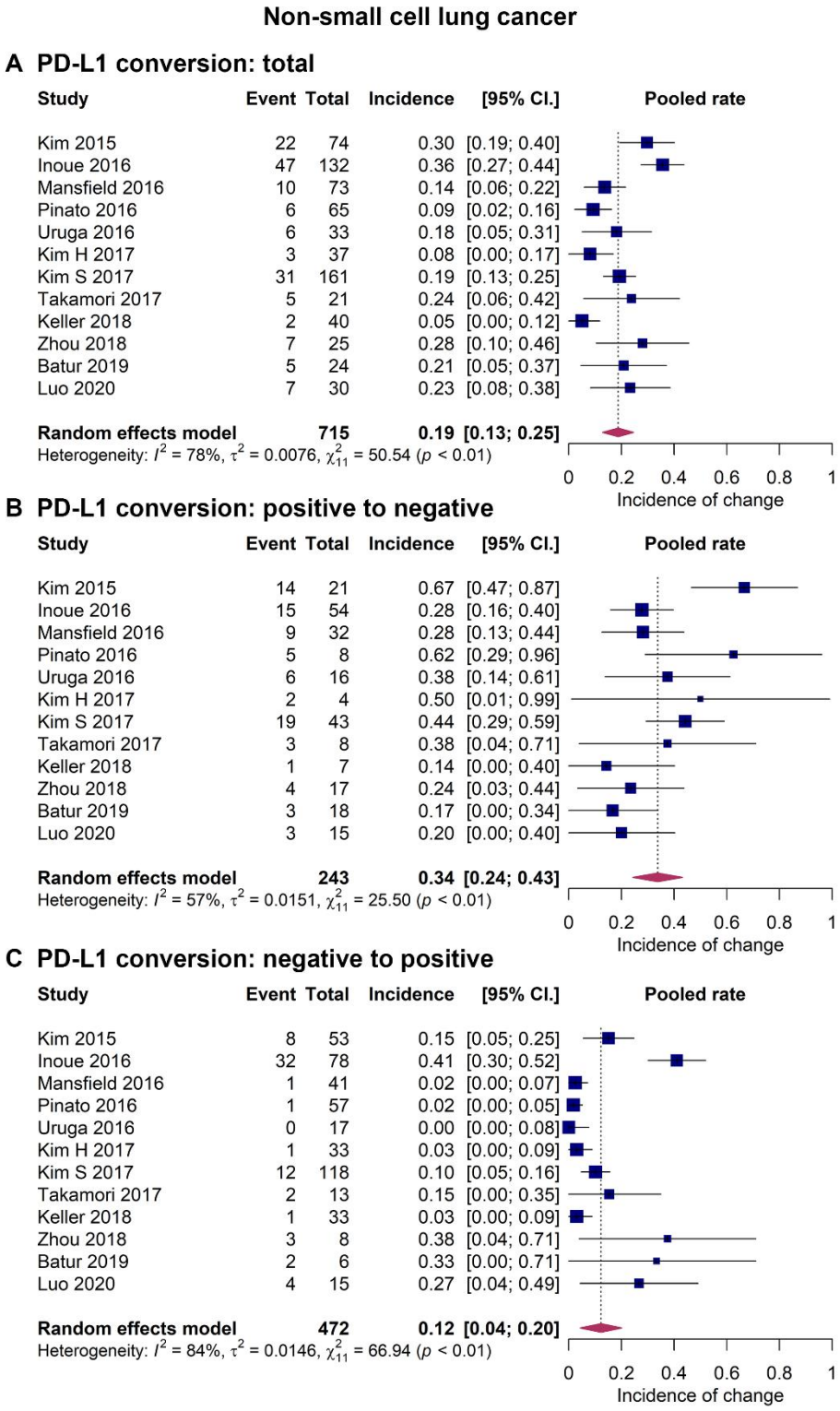


C TIL conversion: low to high

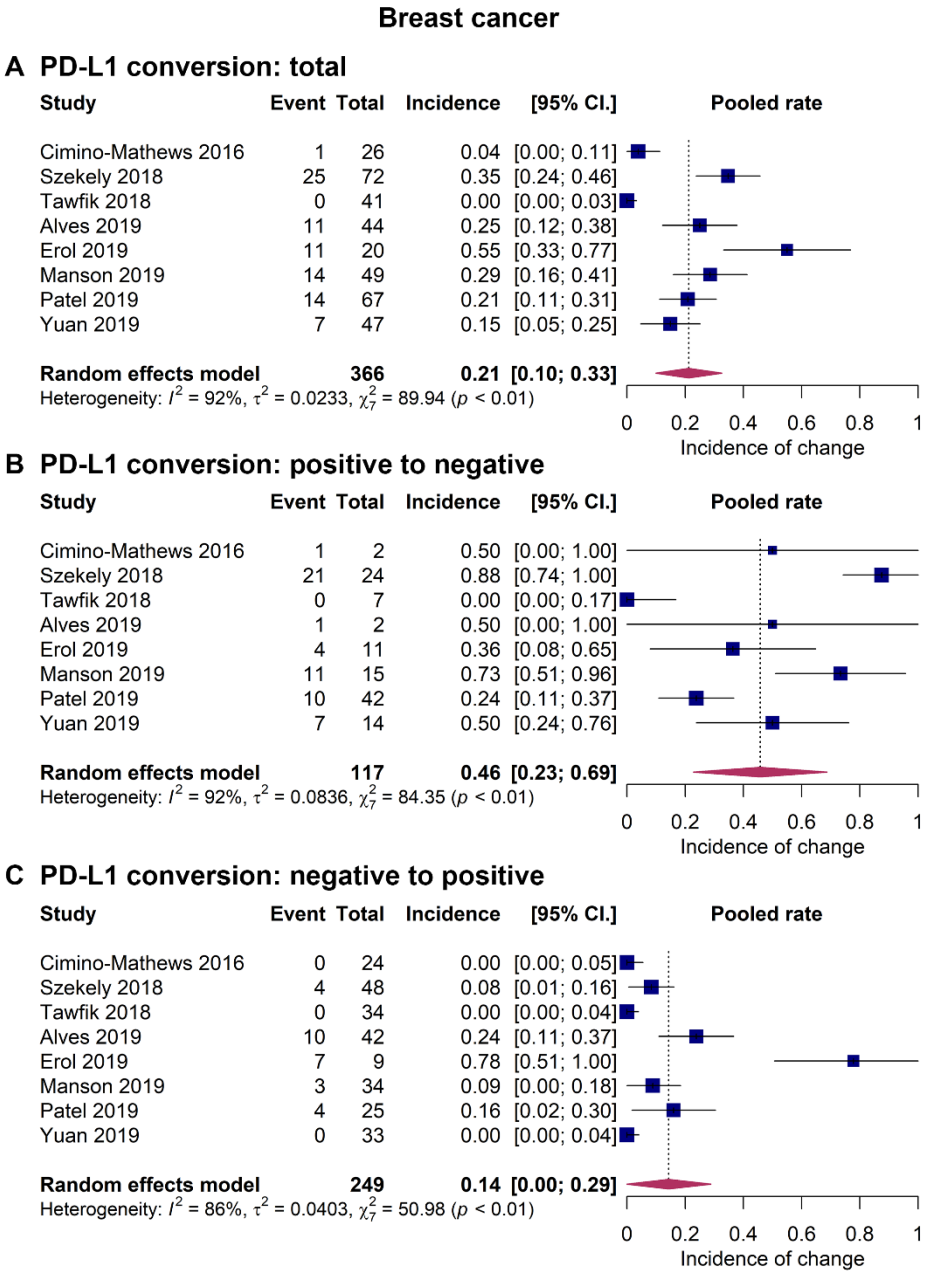
Subgroup	No. of Study	Sample size	Incidence of change	I^2
All patients	**11**	**105**	**0.16 (0.08-0.23)**	**37%**
Cancer type				
Non-small cell lung cancer	3	24	0.34 (0.10-0.52)	28%
Breast cancer	5	52	0.16 (0.06-0.25)	0%
Kidney renal clear cell carcinoma	1	12	0.17 (0.00-0.38)	/
Colorectal cancer	1	14	0.07 (0.00-0.21)	/
Bladder cancer	1	3	0.00 (0.00-0.32)	/
Metastatic site				
Brain metastasis	4	34	0.30 (0.03-0.57)	73%
Liver metastasis	2	17	0.06 (0.00-0.19)	0%
TIL assessment				
HE staining	9	73	0.20 (0.07-0.32)	49%
Bioinformatics	2	32	0.12 (0.01-0.24)	0%
Cut-off value				
5%	5	43	0.21 (0.00-0.42)	69%
10%	4	35	0.17 (0.05-0.29)	0%
Specimen Source				
Whole tissue	9	88	0.17 (0.07-0.28)	49%
Tissue microarray	2	17	0.18 (0.00-0.36)	0%



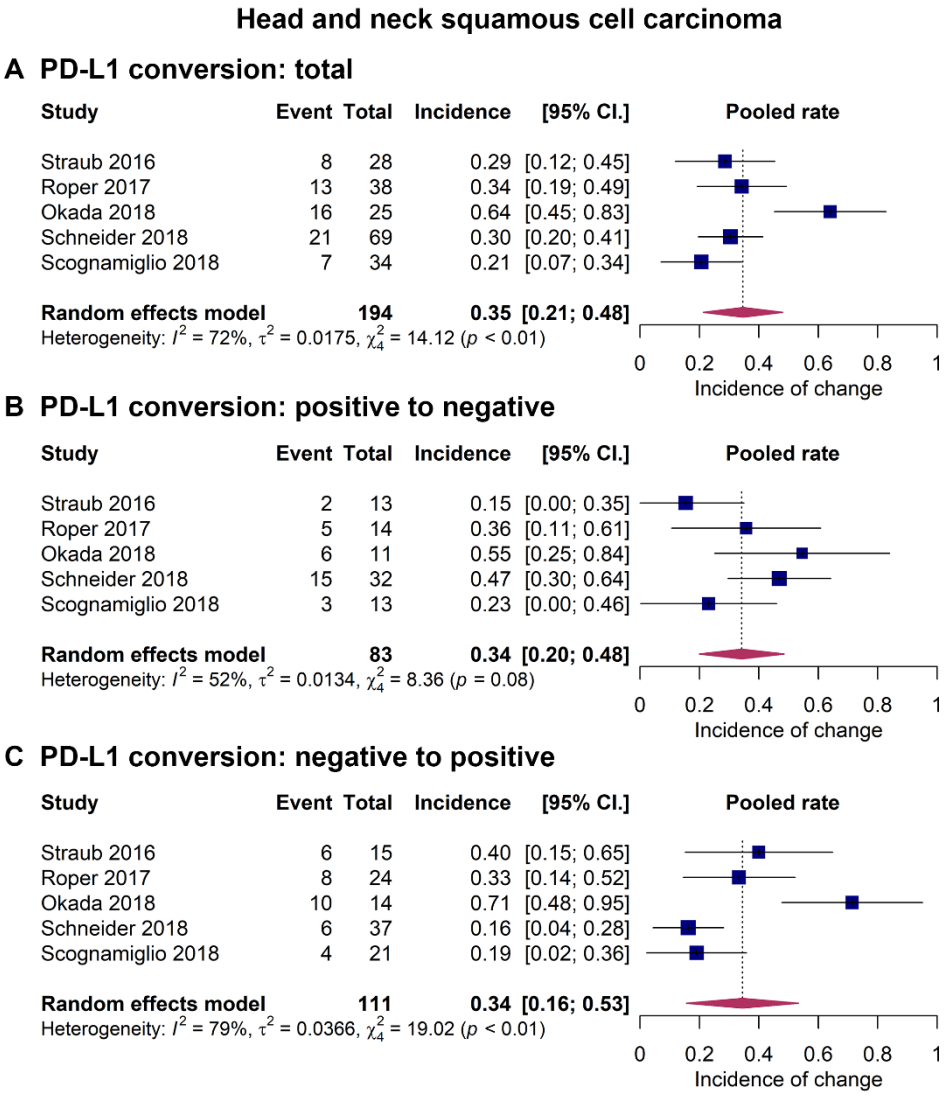
Supplemental Figure S2. Subgroup analysis for the conversion rates of TIL level based on cancer types and metastatic sites. Disaccordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



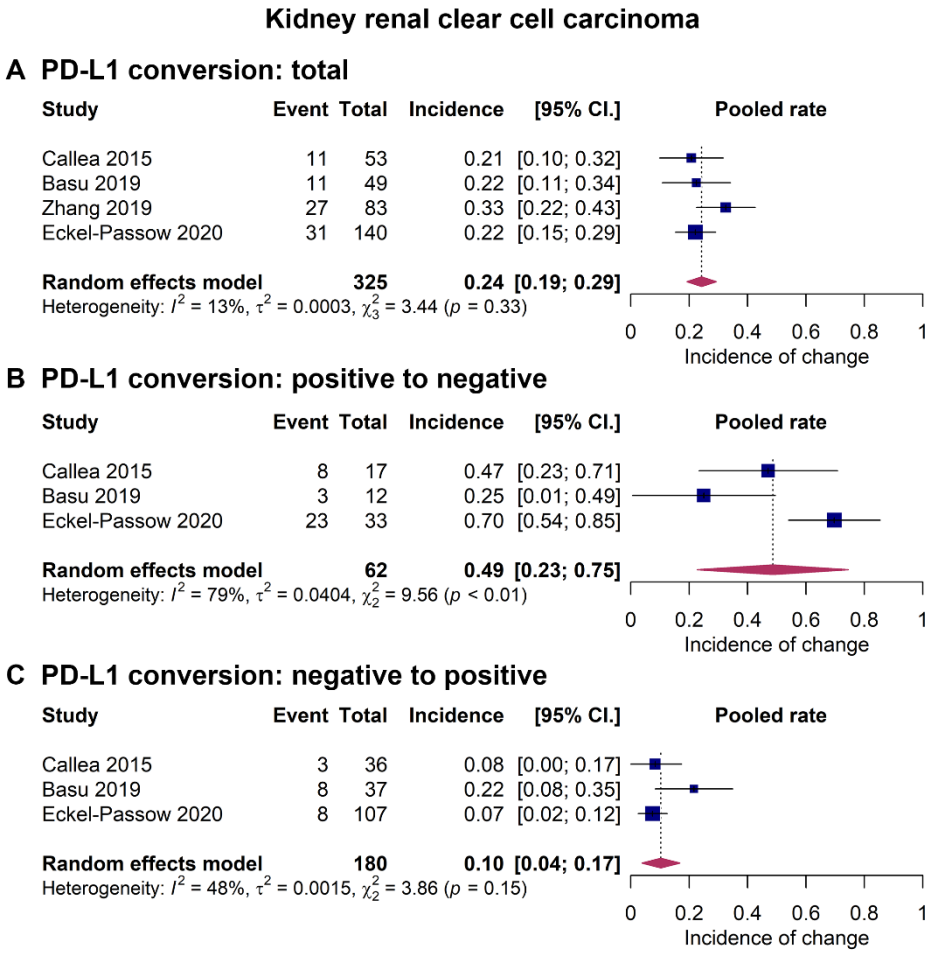
Supplemental Figure S3. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary non-small cell lung cancer and paired metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



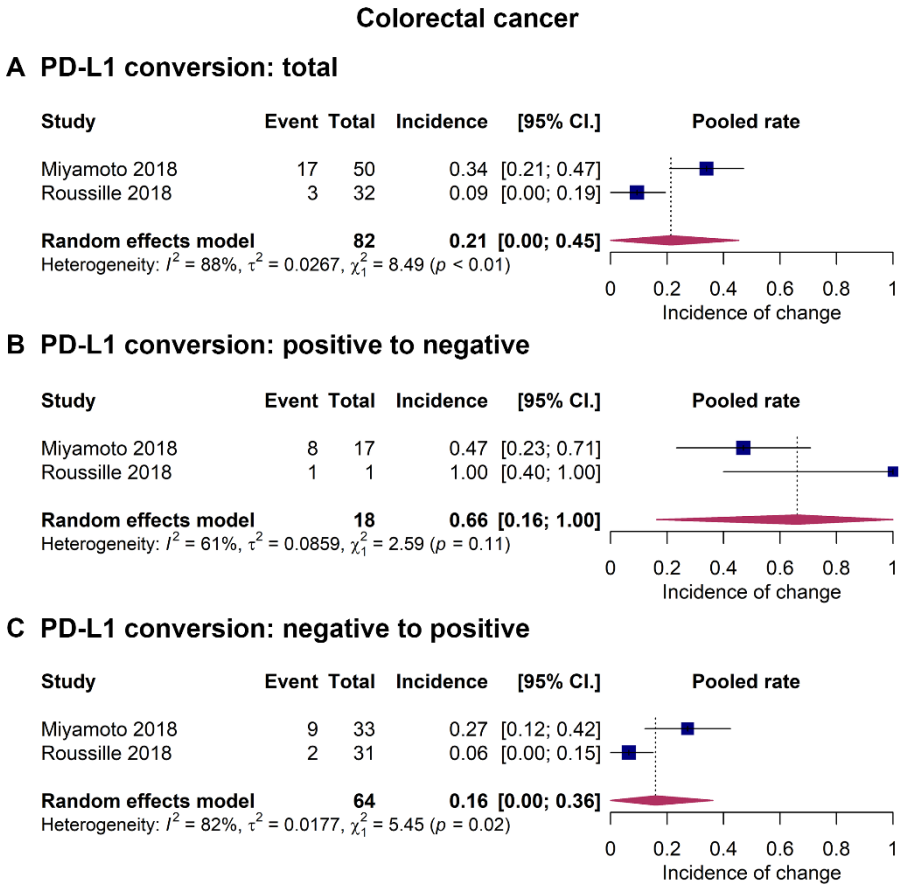
Supplemental Figure S4. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary primary breast cancer and paired metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



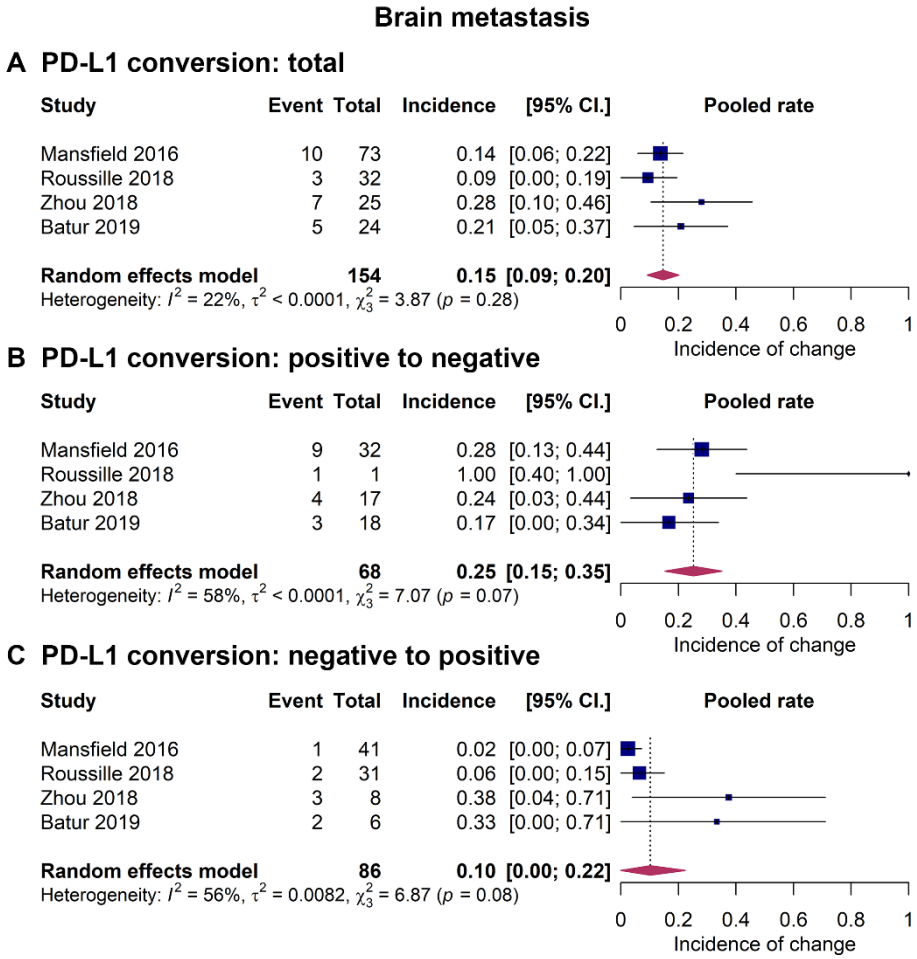
Supplemental Figure S5. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary head and neck squamous cell carcinoma and paired metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



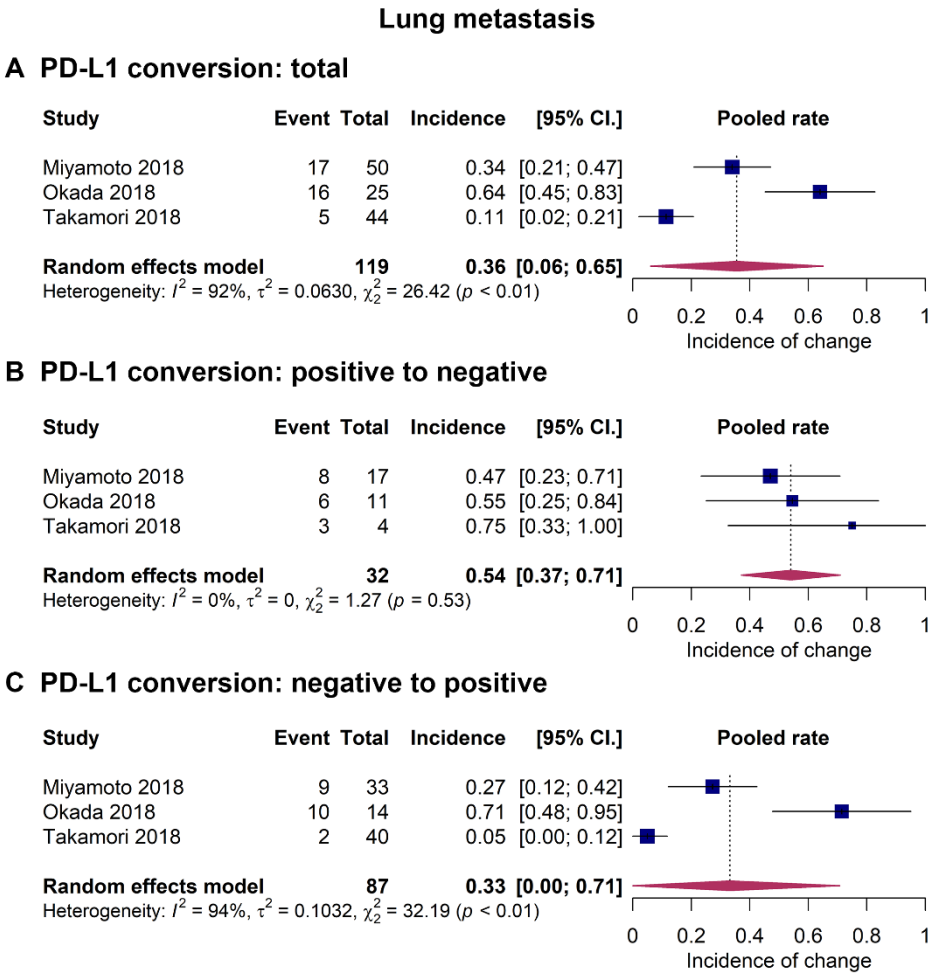
Supplemental Figure S6. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary kidney renal clear cell carcinoma and paired metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



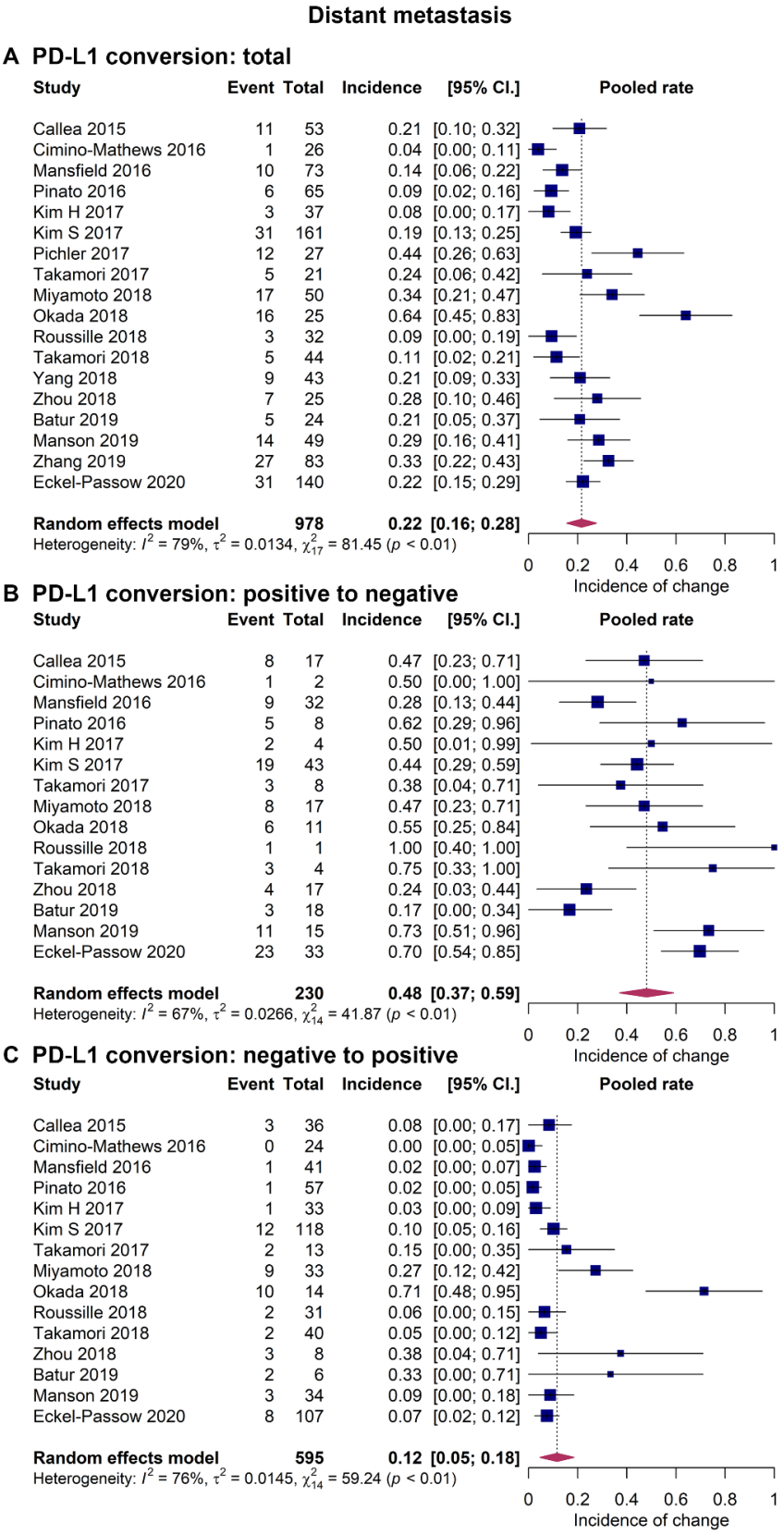
Supplemental Figure S7. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary colorectal cancer and paired metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



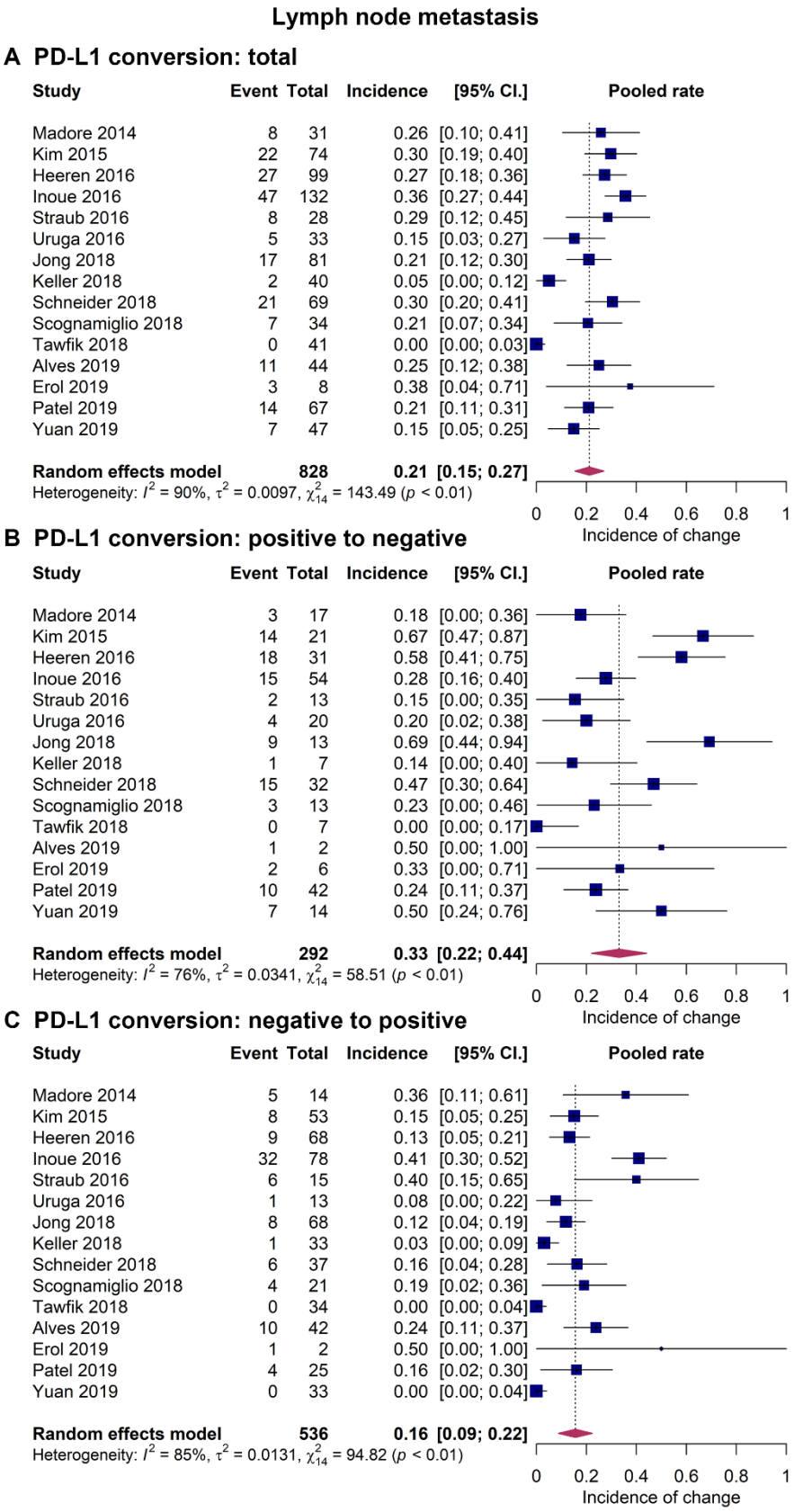
Supplemental Figure S8. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary tumor and paired brain metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



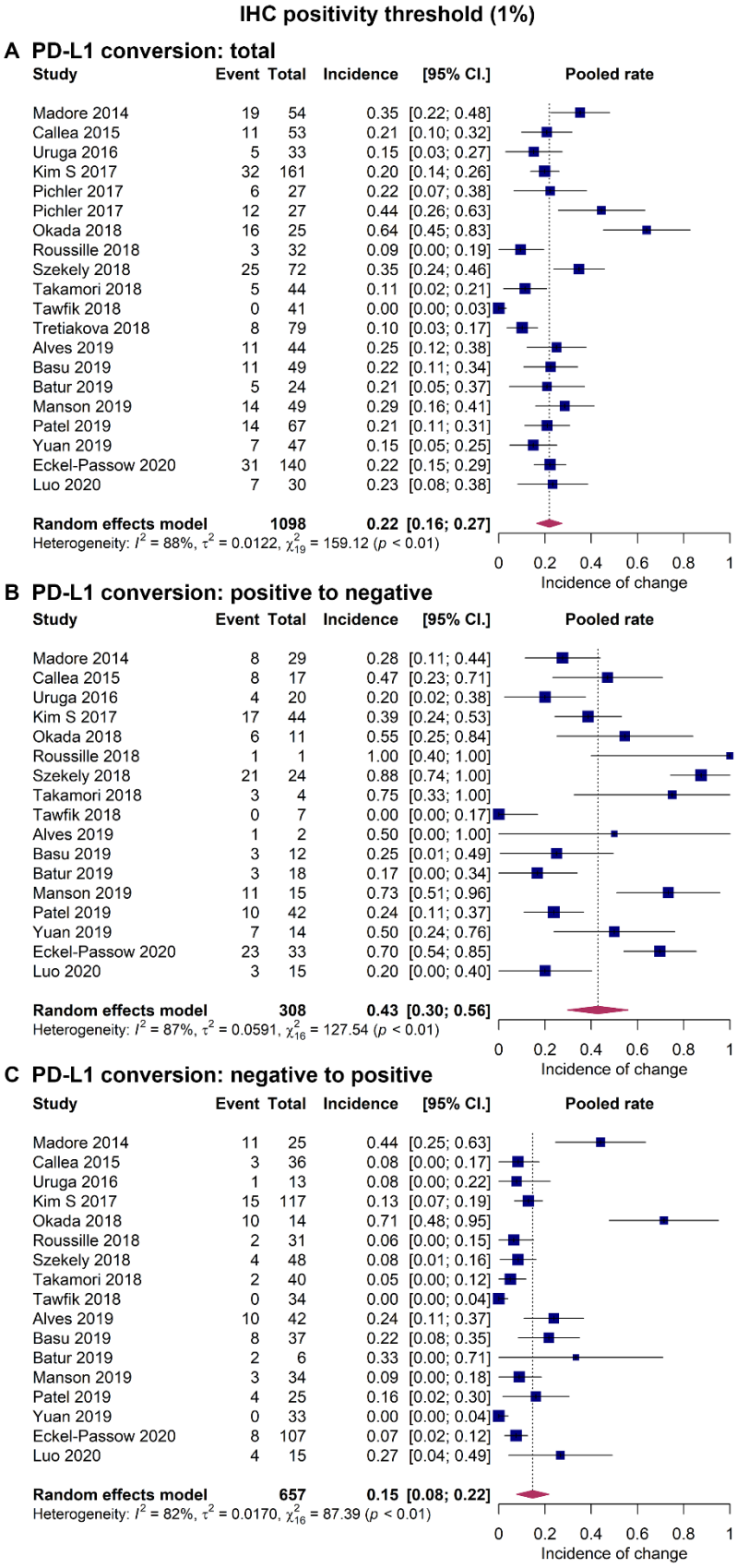
Supplemental Figure S9. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary tumor and paired lung metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



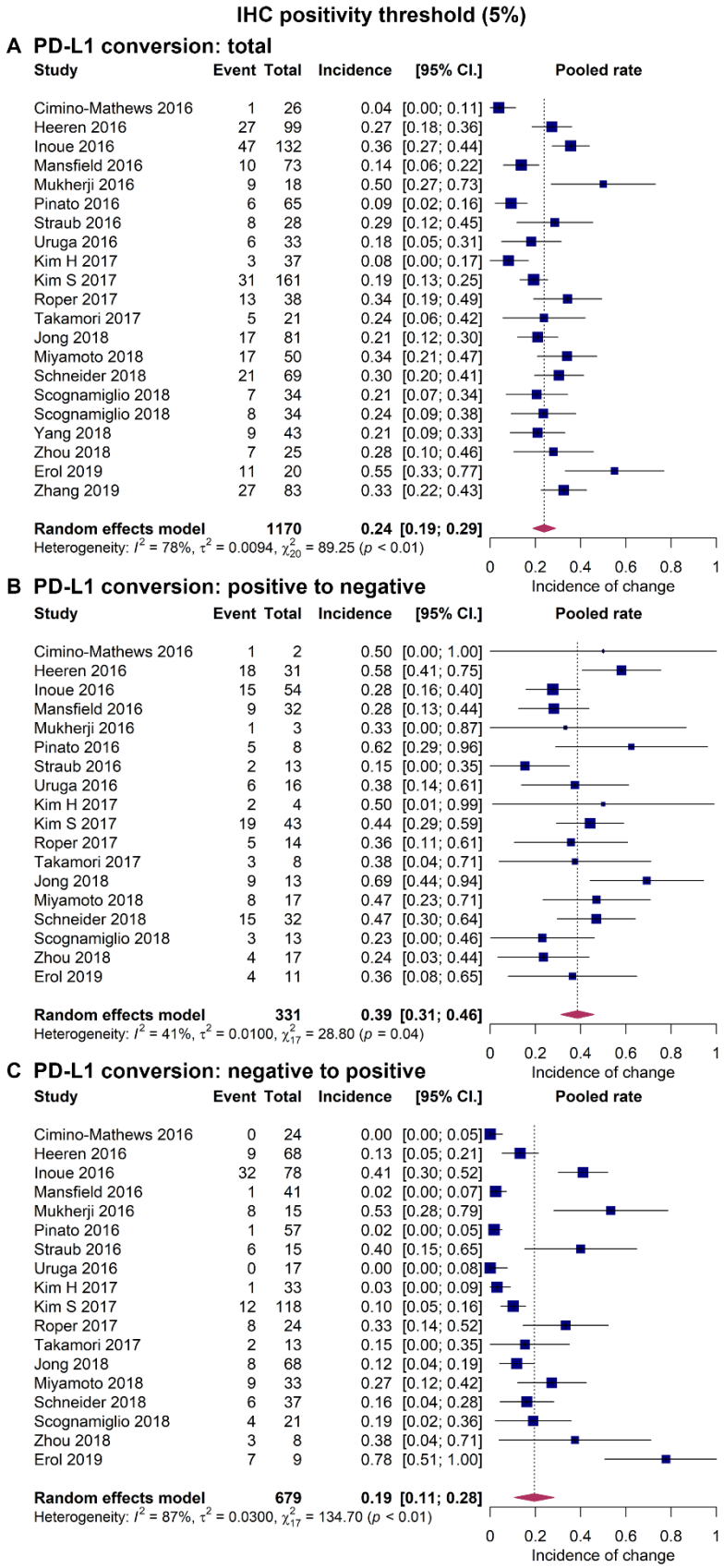
Supplemental Figure S10. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary tumor and paired distant metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



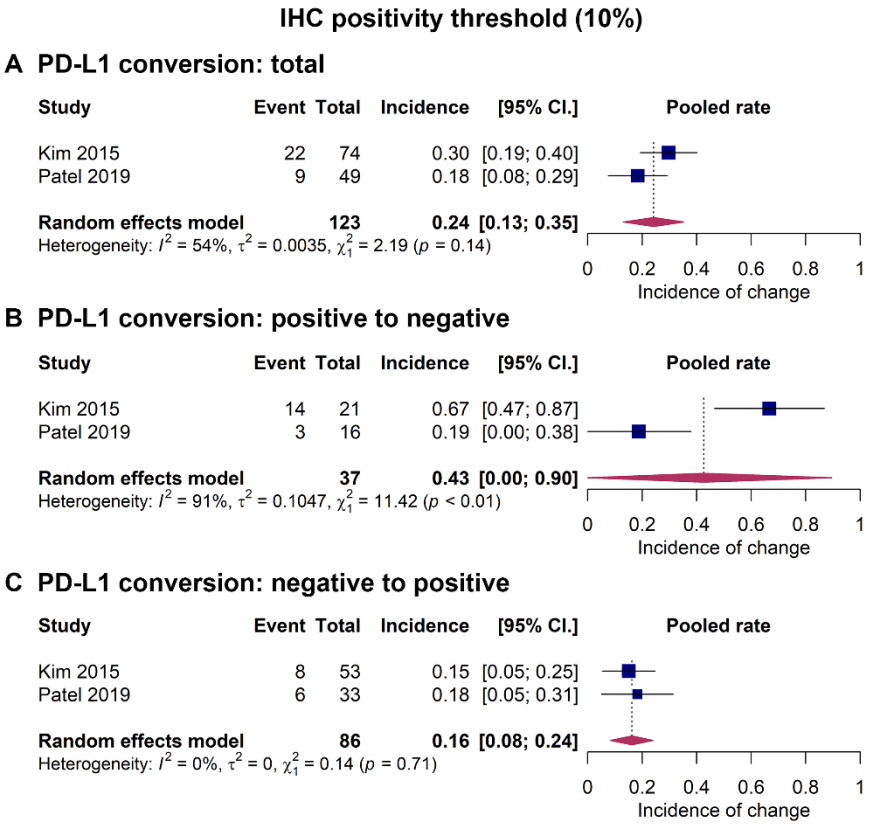
Supplemental Figure S11. Study-specific and pooled estimates for the conversion rates of PD-L1 among primary tumor and paired lymph node metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



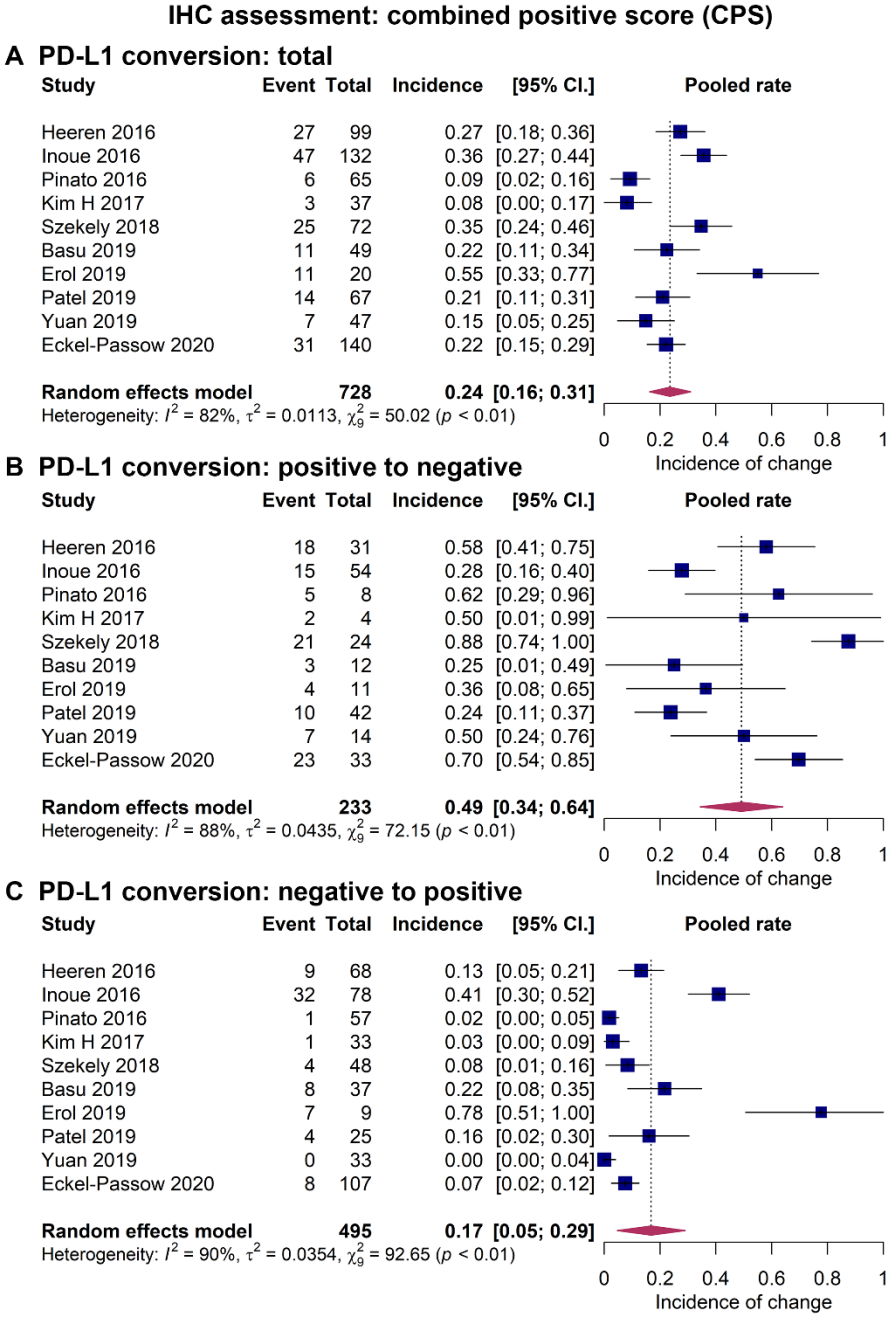
Supplemental Figure S12. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases using 1% as a positive threshold. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



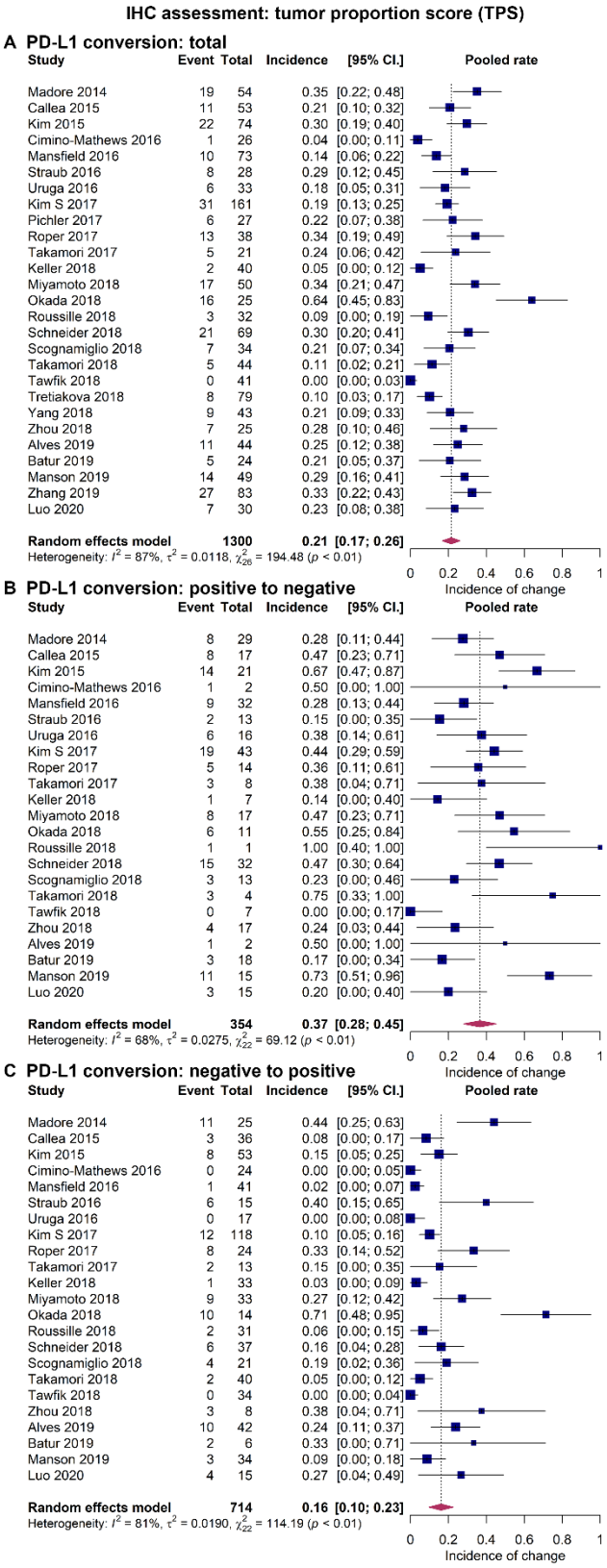
Supplemental Figure S13. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases using 5% as a positive threshold. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



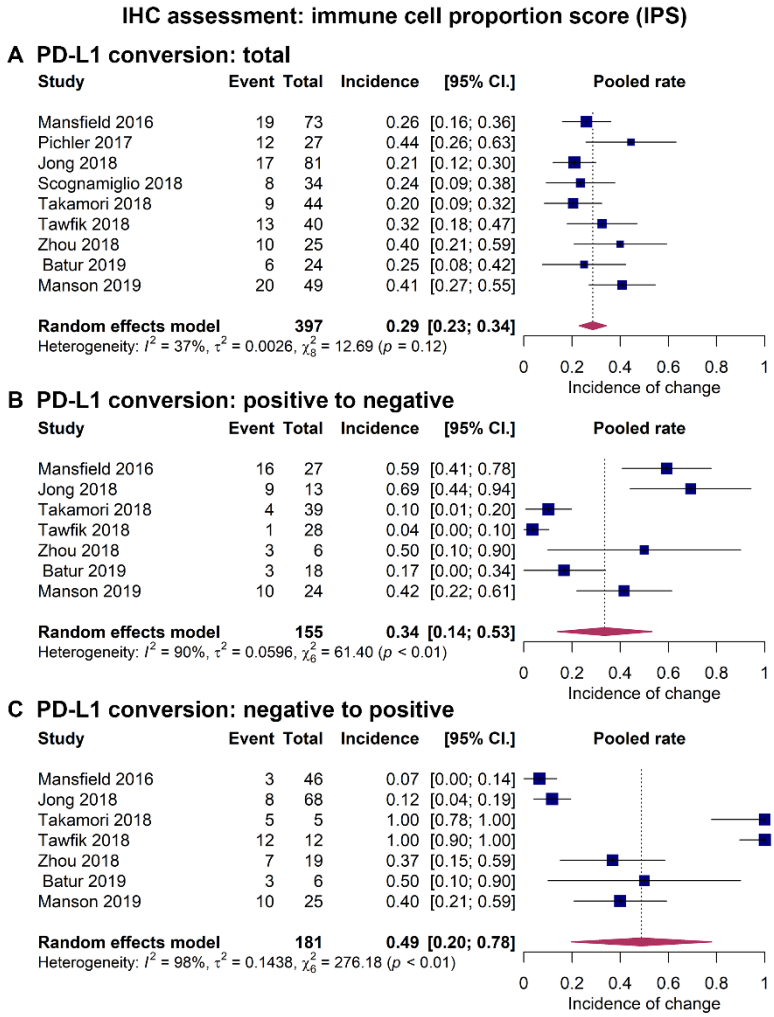
Supplemental Figure S14. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases using 10% as a postivie threshold. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



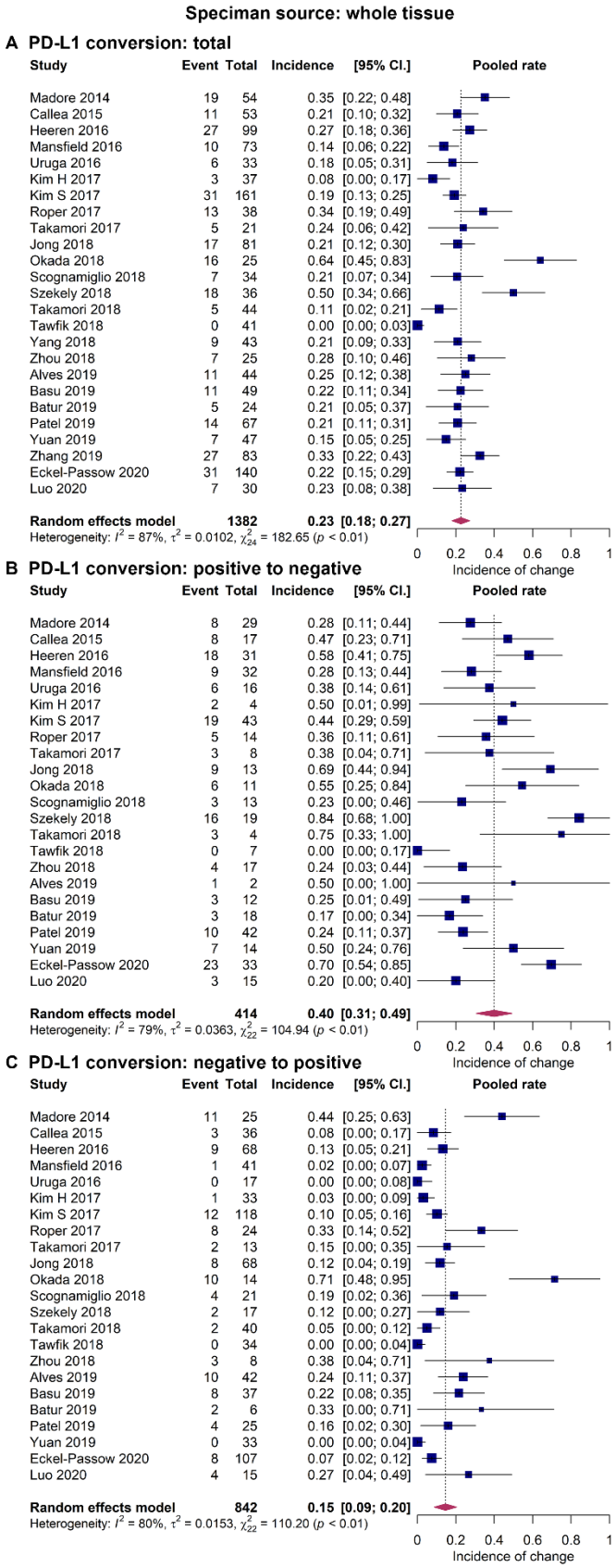
Supplemental Figure S15. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases using CPS assessment . Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



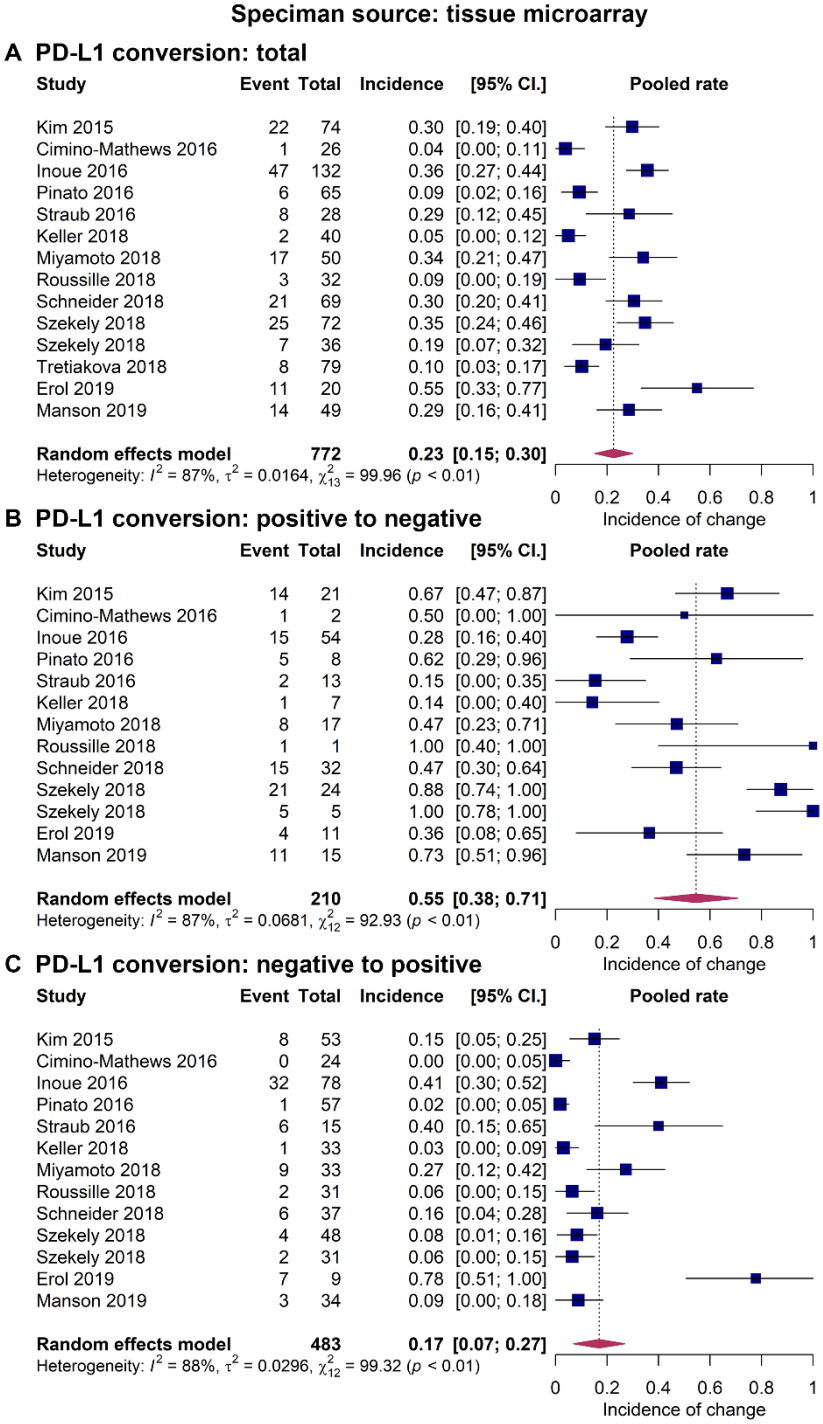
Supplemental Figure S16. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases using TPS assessment. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



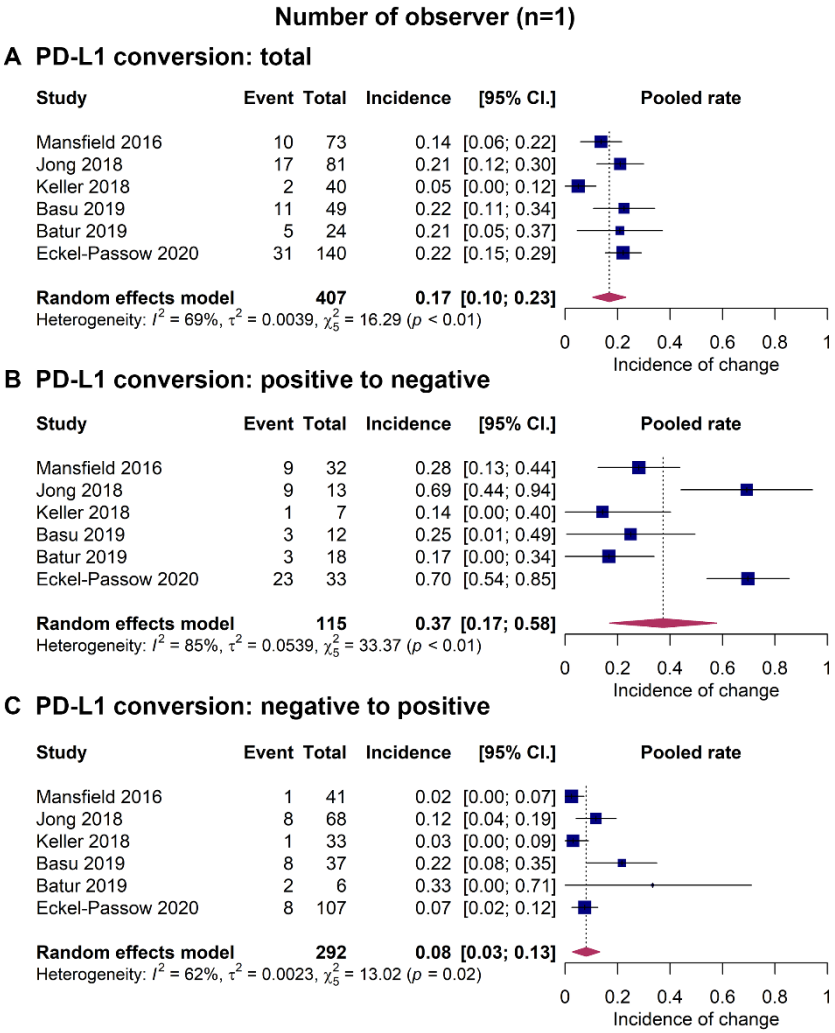
Supplemental Figure S17. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases using IPS assessment. Discordance rate are shown for (A) total conversion, (B) from positive to negative, and (C) from negative to positive.



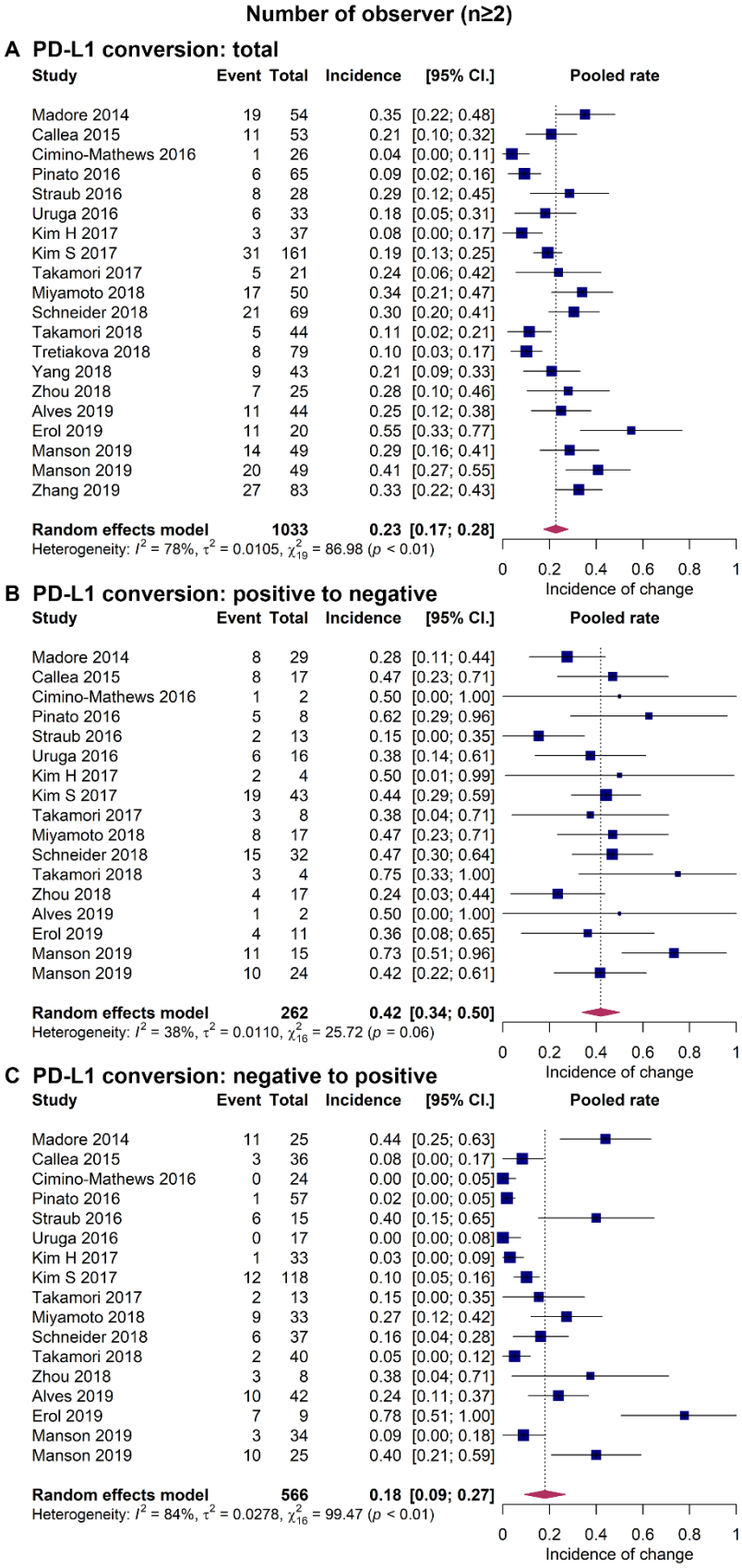
Supplemental Figure S18. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases assessed by whole tissue IHC. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



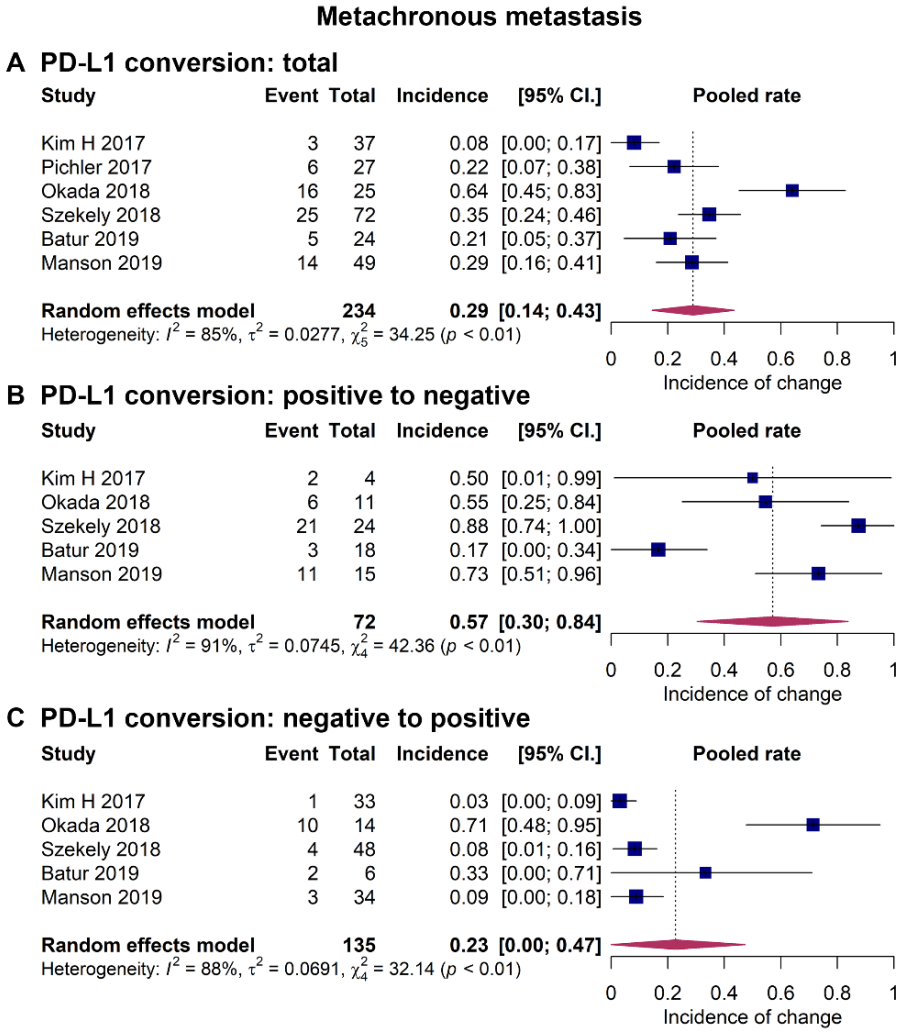
Supplemental Figure S19. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases assessed by tissue microarray IHC. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



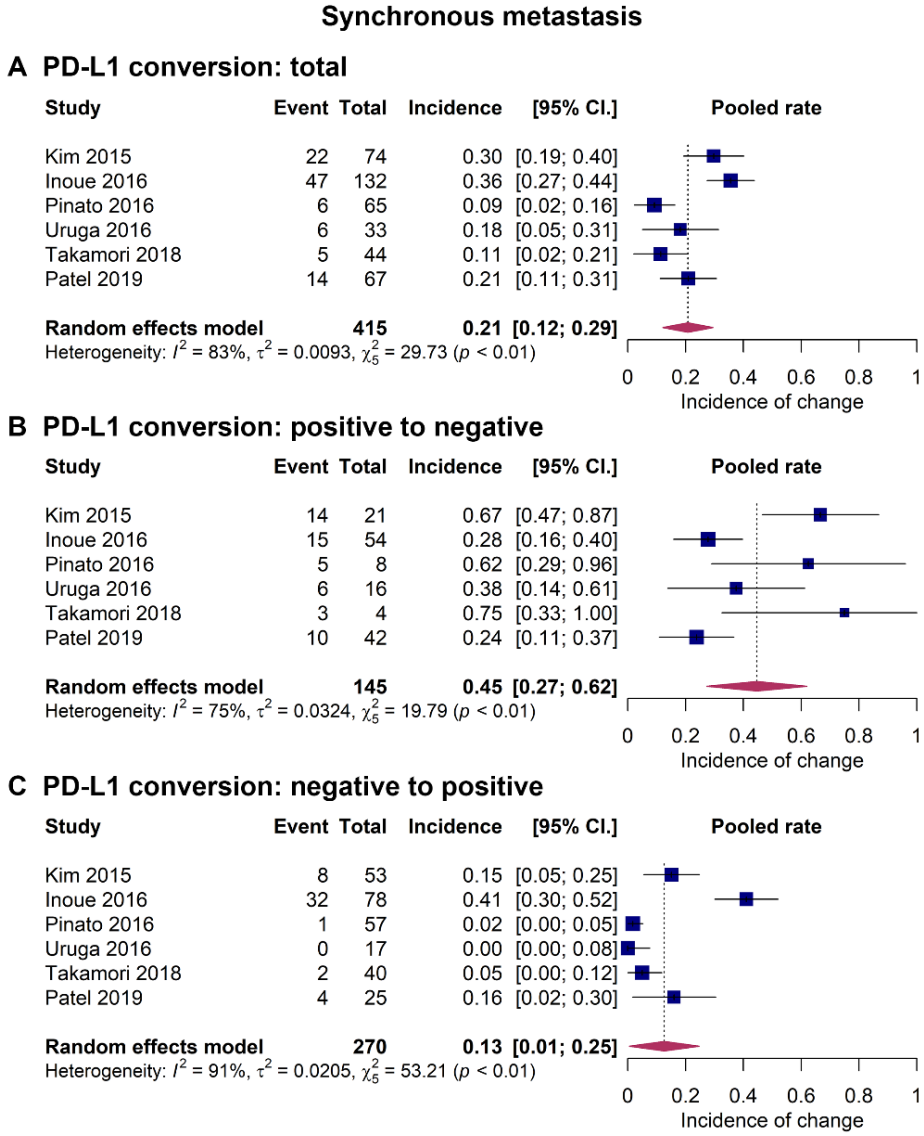
Supplemental Figure S20. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases assessed by one observer. Discordance rateS are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



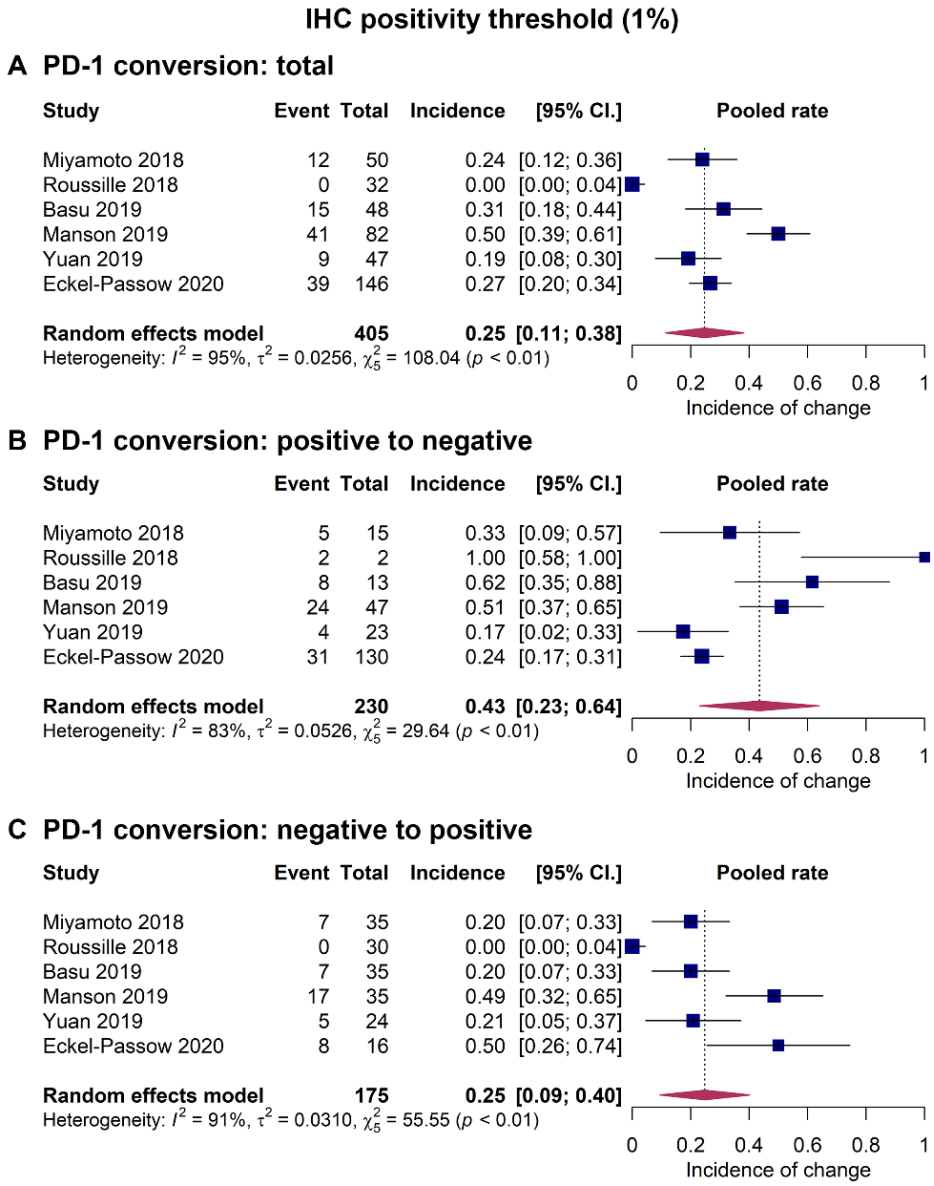
Supplemental Figure S21. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metastases assessed by two or more observers. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



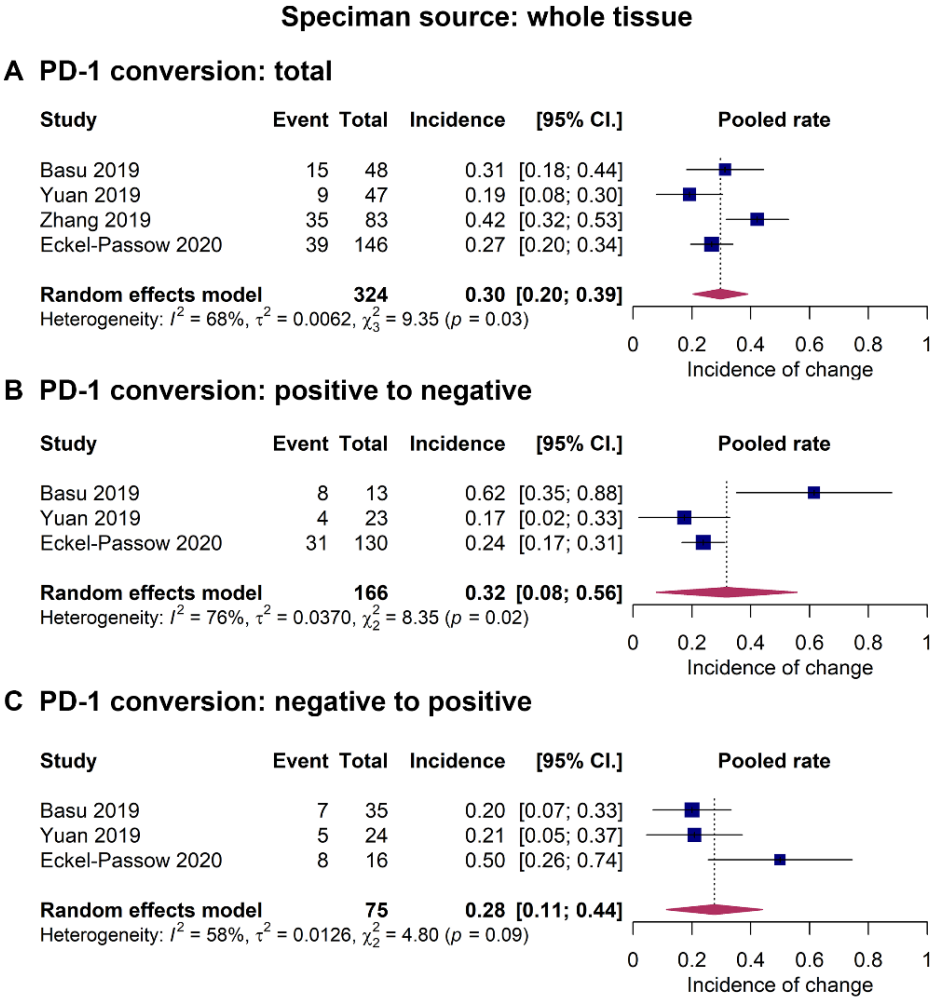
Supplemental Figure S22. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired metachronous metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



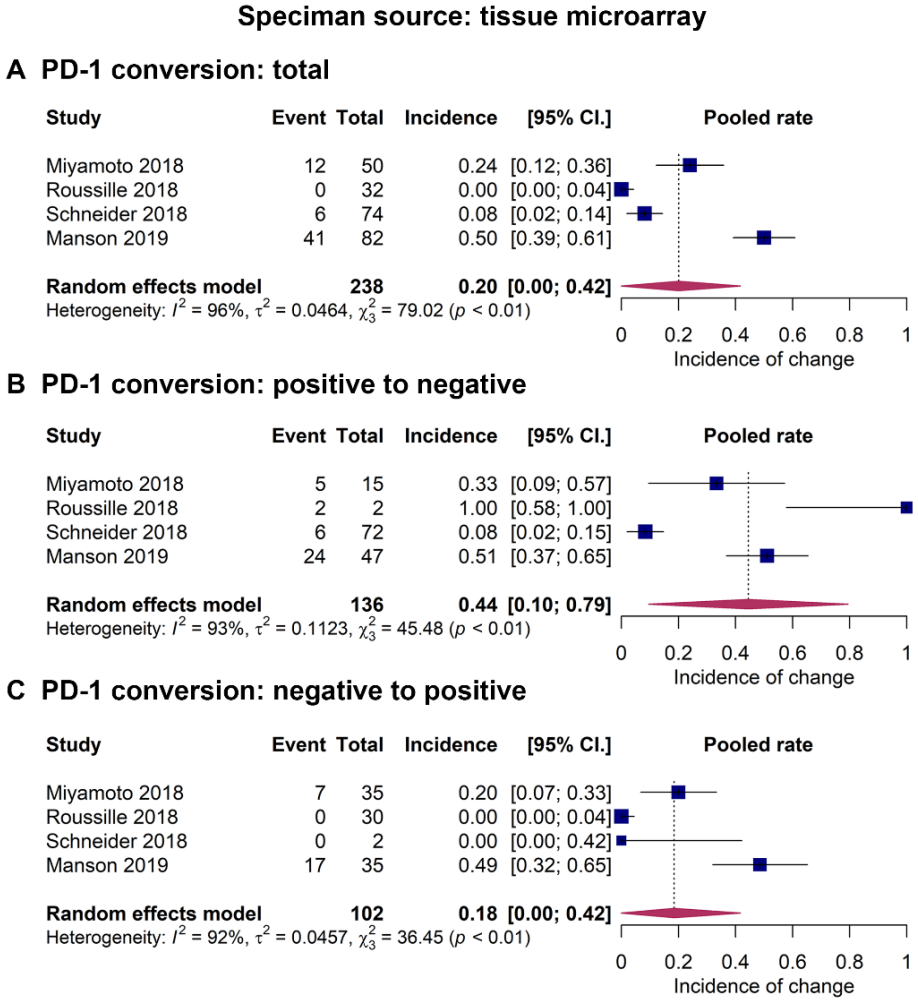
Supplemental Figure S23. Study-specific and pooled estimates for PD-L1 conversion rates among primary tumor and paired synchronous metastases. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



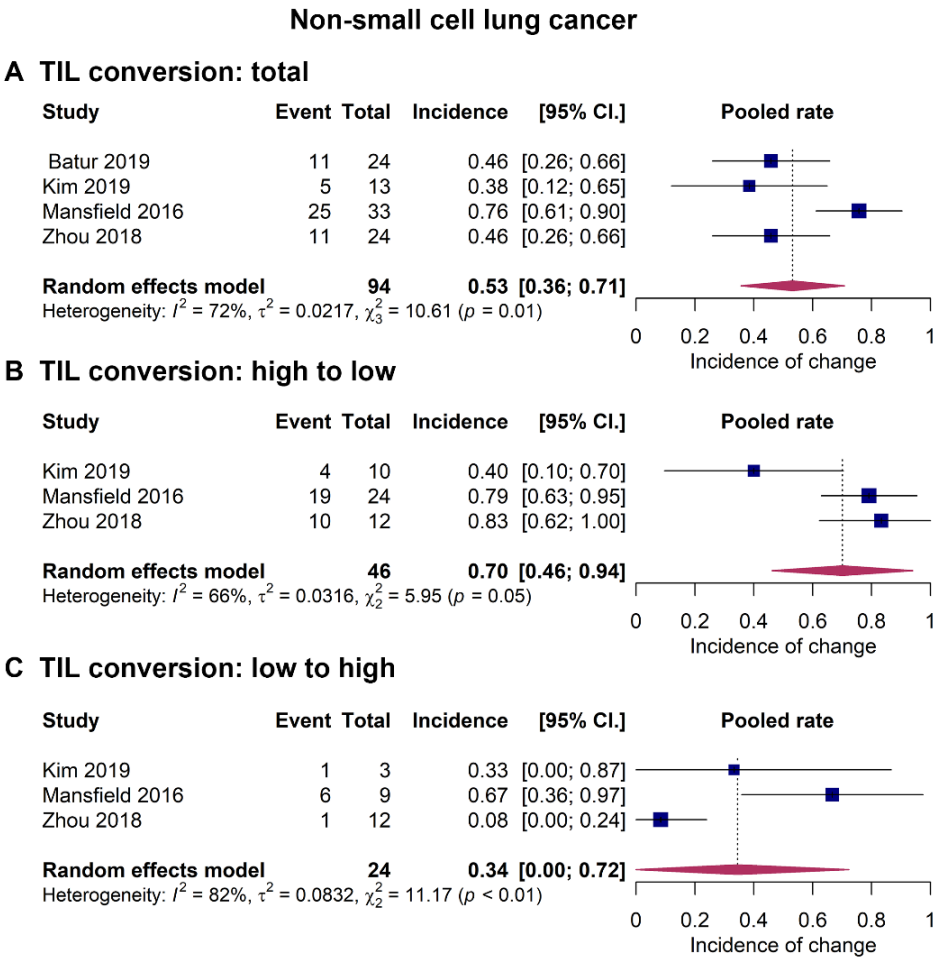
Supplemental Figure S24. Study-specific and pooled estimates for PD-1 conversion rates among primary tumor and paired metastases using 1% as a positive threshold. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



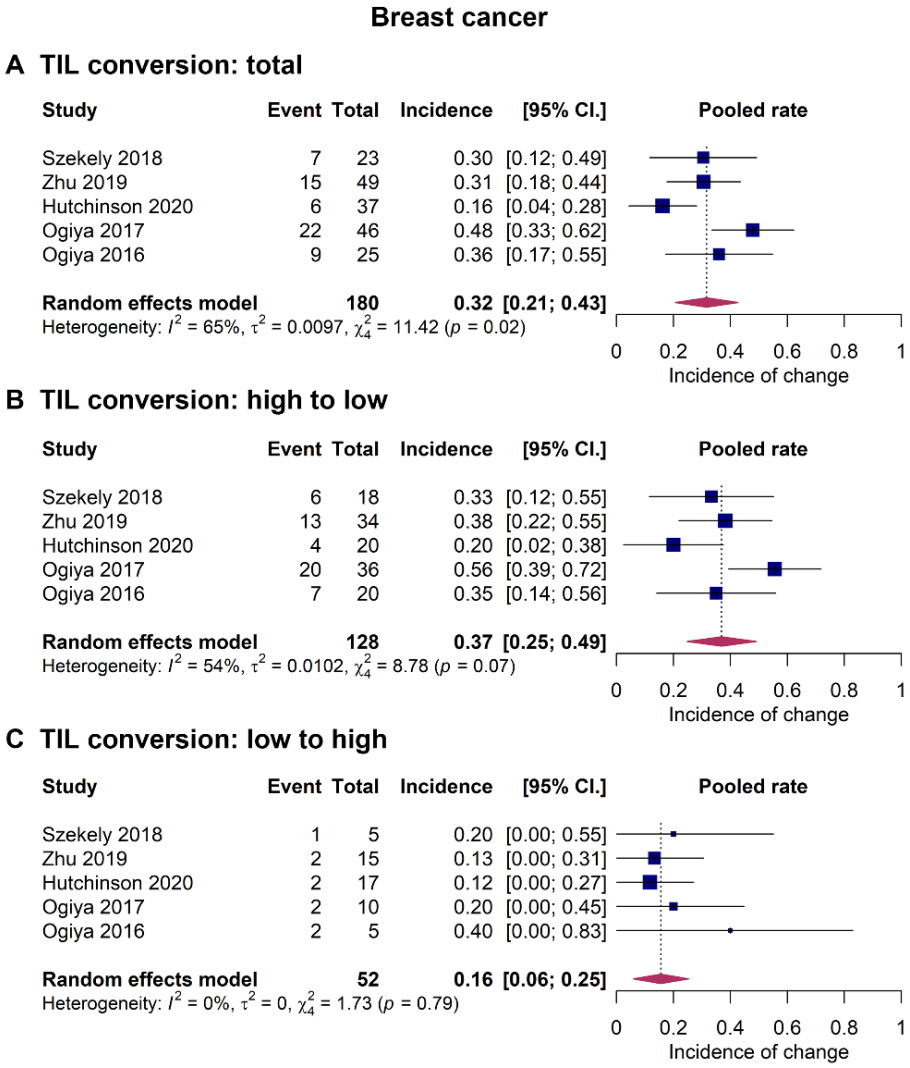
Supplemental Figure S25. Study-specific and pooled estimates for PD-1 conversion rates among primary tumor and paired metastases assessed by whole tissue IHC. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



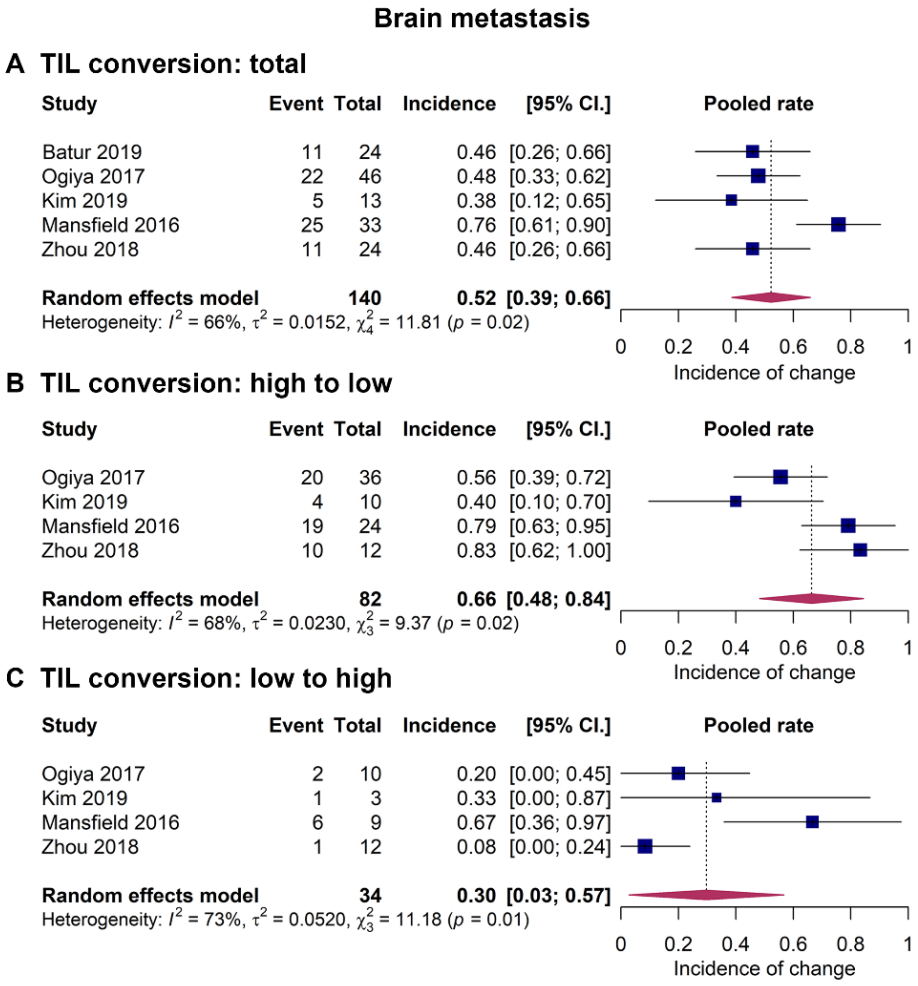
Supplemental Figure S26. Study-specific and pooled estimates for PD-1 conversion rates among primary tumor and paired metastases assessed by tissue microarray IHC. Discordance rates are shown for (A) total, (B) from positive to negative, and (C) from negative to positive conversion.



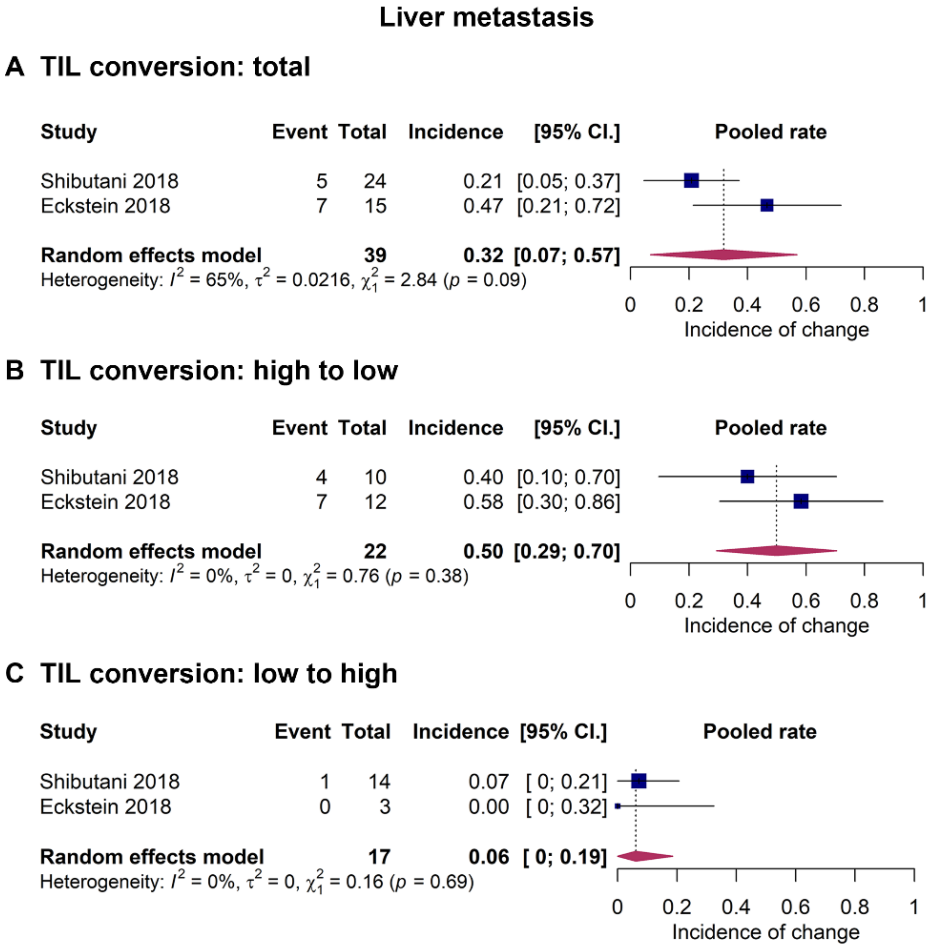
Supplemental Figure S27. Study-specific and pooled estimates for the conversion rates of TIL level among primary non-small cell lung cancer and paired metastases. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



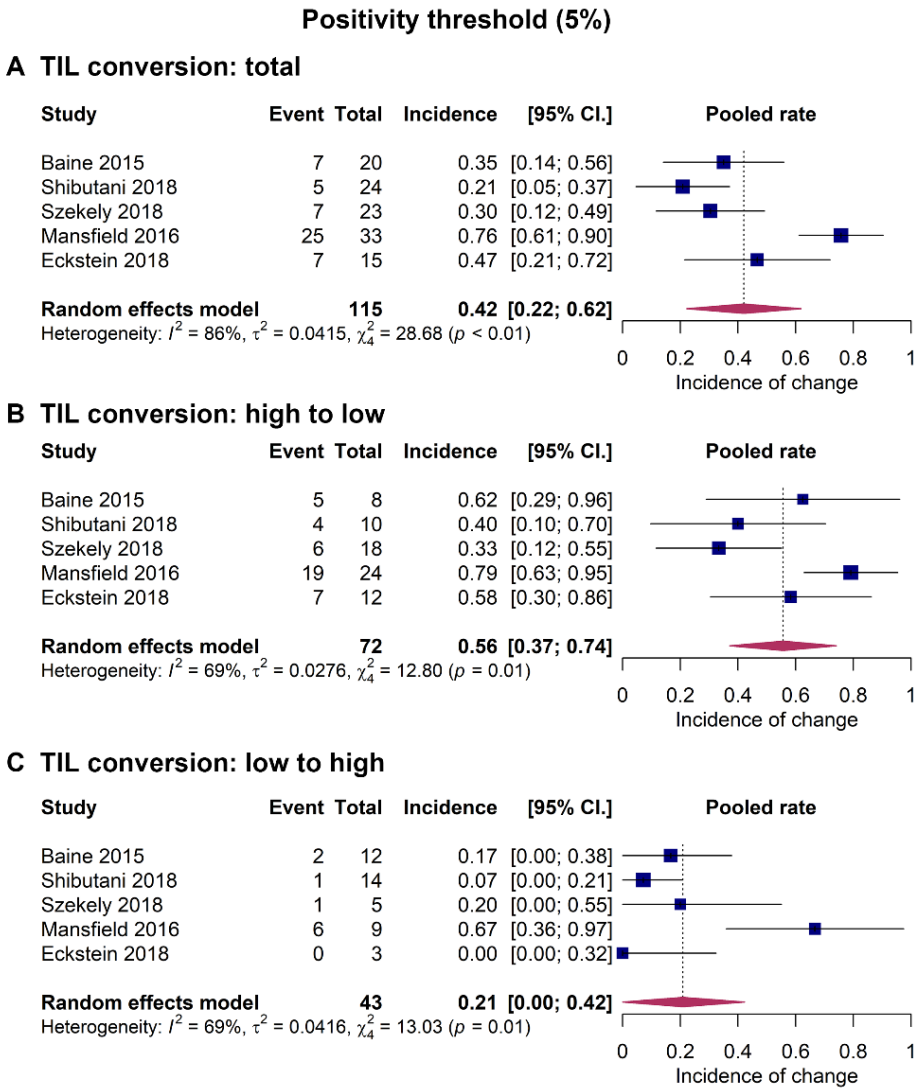
Supplemental Figure S28. Study-specific and pooled estimates for the conversion rates of TIL level among primary breast cancer and paired metastases. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



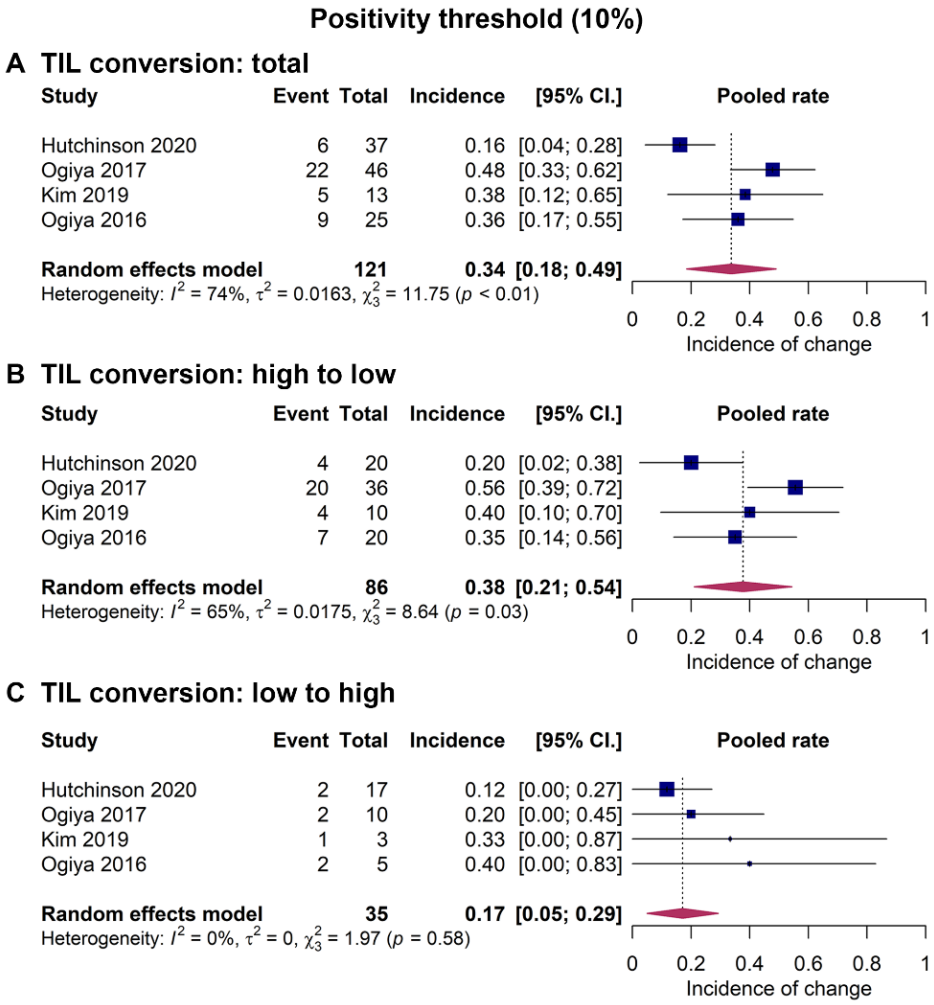
Supplemental Figure S29. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired brain metastases. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



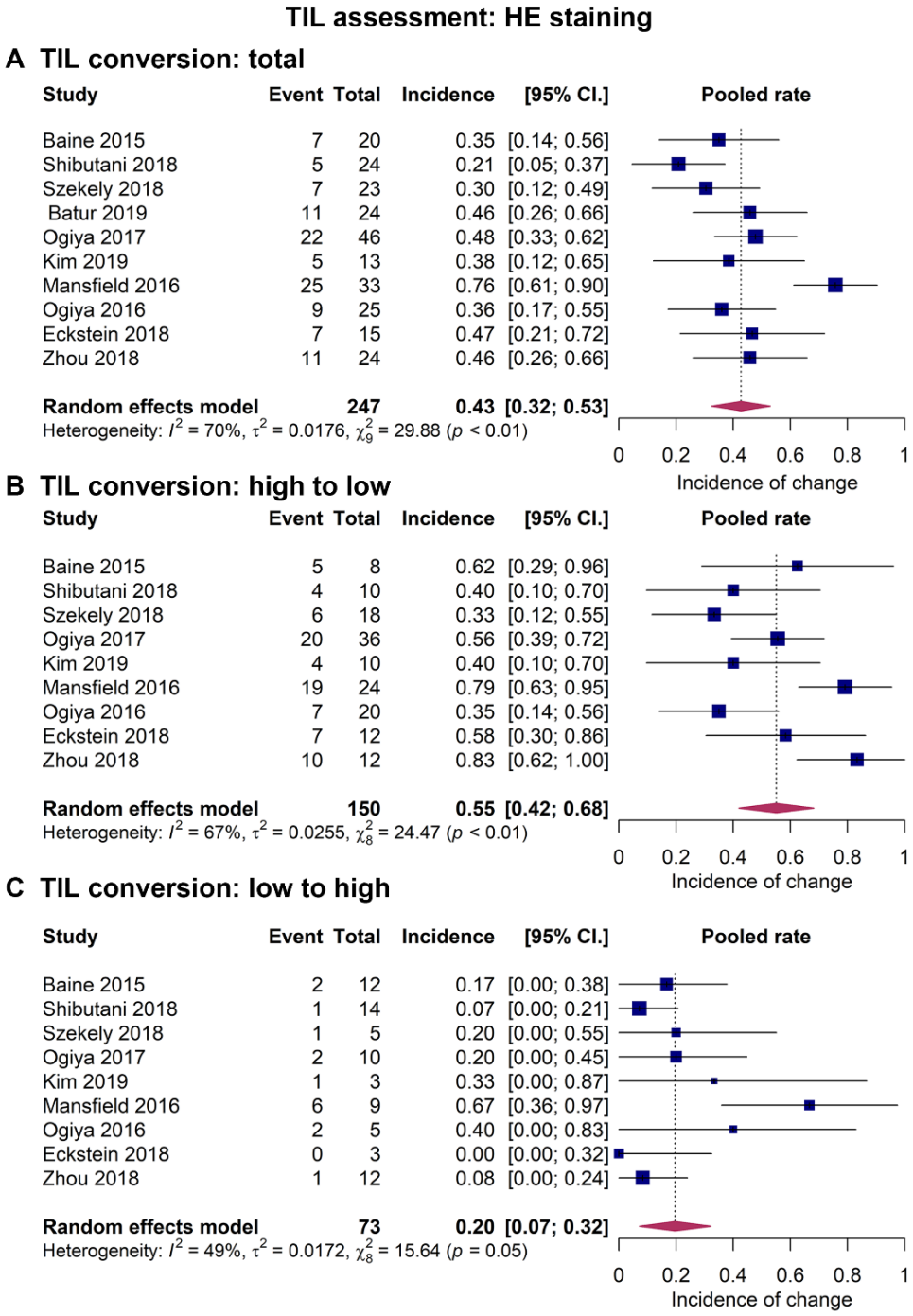
Supplemental Figure S30. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired liver metastases. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



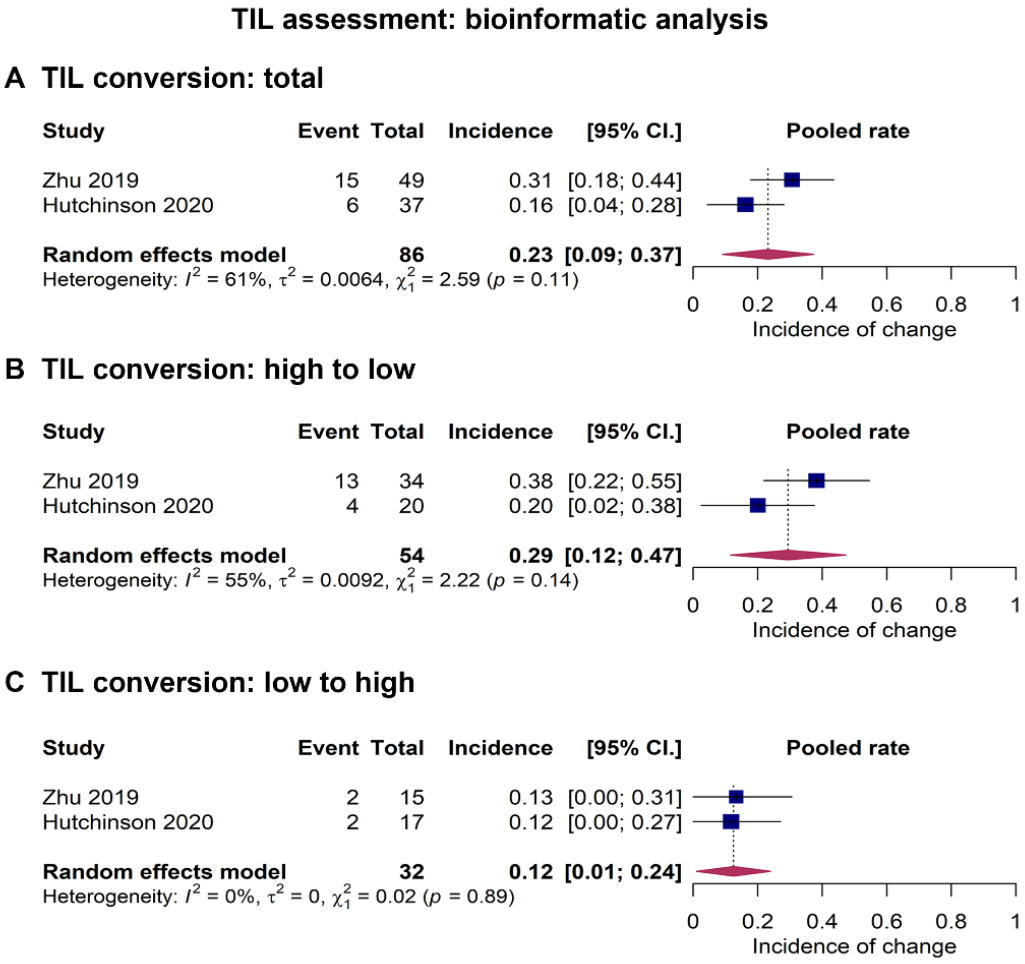
Supplemental Figure S31. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired metastases using 5% as a positive threshold. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



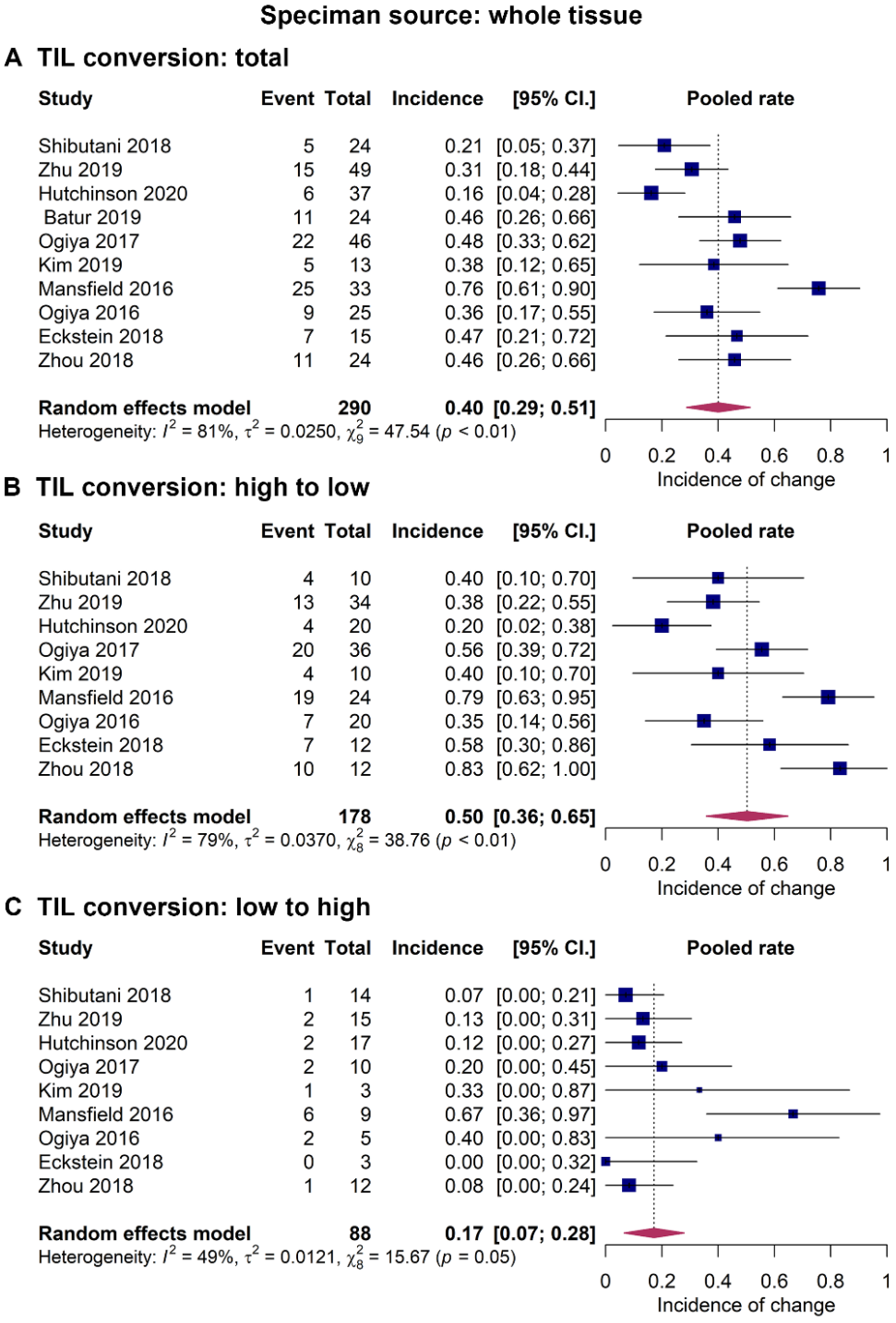
Supplemental Figure S32. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired metastases using 10% as a postive threshold. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



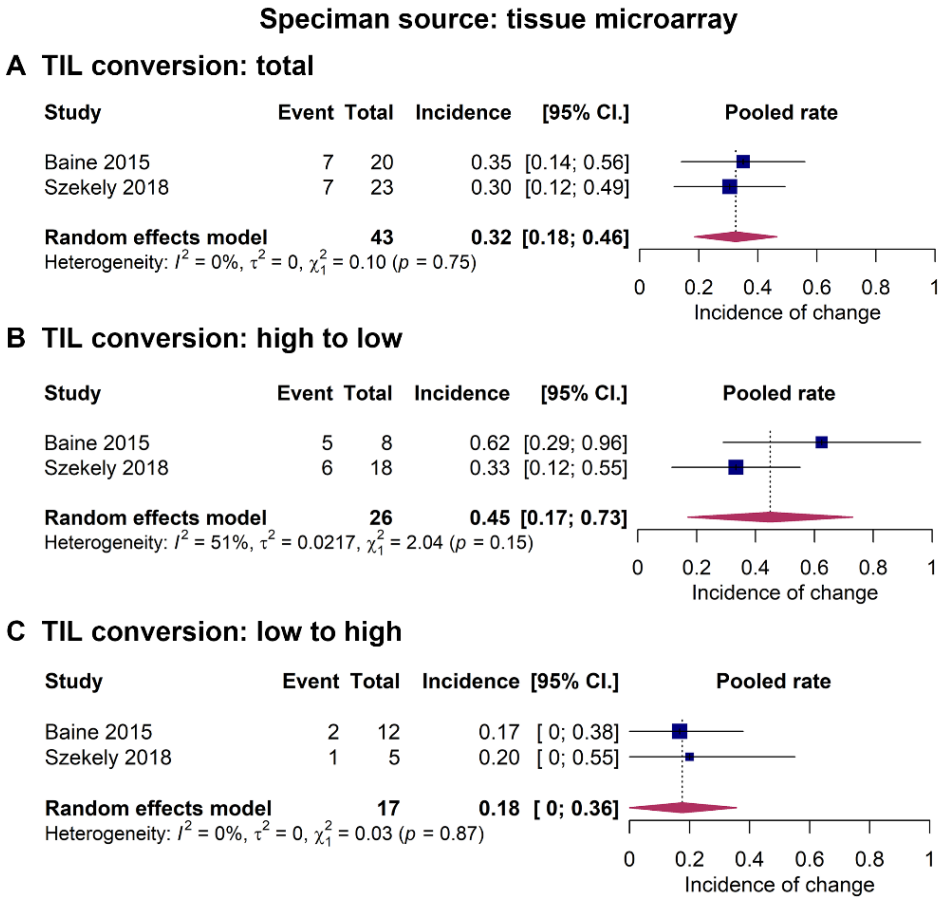
Supplemental Figure S33. Study-specific and pooled estimates for the conversion rates among primary tumor and paired metastases using HE staining method. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



Supplemental Figure S34. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired metastases using bioinformatic analysis method. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



Supplemental Figure S35. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired metastases assessed by whole tissue IHC. Discordance rates are shown for (A) total, (B) from high to low, and (C) from low to high conversion.



Supplemental Figure S36. Study-specific and pooled estimates for the conversion rates of TIL level among primary tumor and paired metastases assessed by tissue microarray IHC. Discordance rates are shown for (A) total conversion, (B) from high to low, and (C) from low to high.

Supplementary Methods

Search strategy

Pubmed

((((((((((((neoplas*[Title/Abstract]) OR tumor[Title/Abstract]) OR tumors[Title/Abstract]) OR tumour*[Title/Abstract]) OR cancer*[Title/Abstract]) OR carcinom*[Title/Abstract]) OR malignan*[Title/Abstract]) OR oncolog*[Title/Abstract])) OR "Neoplasms"[Mesh])) AND (("Neoplasm Metastasis"[Mesh]) OR ((((metasta*[Title/Abstract]) OR progressi*[Title/Abstract]) OR disseminat*[Title/Abstract]) OR pair*[Title/Abstract]))) AND ((((((((((((((immune biomarker[Title/Abstract]) OR (anti pd-1[Title/Abstract]) OR anti pd-l1[Title/Abstract]) OR anti pd-l2[Title/Abstract]) OR pd-1[Title/Abstract]) OR pd-l1[Title/Abstract]) OR pd-l2[Title/Abstract]) OR pd1[Title/Abstract]) OR pdl1[Title/Abstract]) OR pdl2[Title/Abstract]) OR programmed cell death 1 receptor[Title/Abstract]) OR programmed death ligand[Title/Abstract]) OR "Lymphocytes, Tumor-Infiltrating"[Mesh]) OR "Programmed Cell Death 1 Ligand 2 Protein"[Mesh]) OR "Programmed Cell Death 1 Receptor"[Mesh]) OR "B7-H1 Antigen"[Mesh]))) AND ((((((((((((((((((microsatellite instability [Title/Abstract]) OR mutational burden [Title/Abstract]) OR alteration[Title/Abstract]) OR altered[Title/Abstract]) OR discordan*[Title/Abstract]) OR concordan*[Title/Abstract]) OR chang*[Title/Abstract]) OR changing[Title/Abstract]) OR conver*[Title/Abstract]) OR congruen*[Title/Abstract]) OR incongruen*[Title/Abstract]) OR dyscongruen*[Title/Abstract]) OR discrepan*[Title/Abstract]) OR diversi*[Title/Abstract]) OR difference*[Title/Abstract]) OR differing[Title/Abstract]) OR different[Title/Abstract]) OR comparison[Title/Abstract]) OR heterogen*[Title/Abstract])

Cochrane library

#1 MeSH descriptor: [Neoplasm Metastasis] explode all trees

#2 MeSH descriptor: [Neoplasms] explode all trees

#3 MeSH descriptor: [B7-H1 Antigen] explode all trees

#4 MeSH descriptor: [Programmed Cell Death 1 Receptor] explode all trees

#5 MeSH descriptor: [Lymphocytes, Tumor-Infiltrating] explode all trees

#6 (pd-l1):ti,ab,kw OR (pdl1):ti,ab,kw OR (pd1):ti,ab,kw OR (pd-1):ti,ab,kw OR (pd-l2):ti,ab,kw (Word variations have been searched)

#7 (alteration):ti,ab,kw OR (altered):ti,ab,kw OR (discordan*):ti,ab,kw OR (concordan*):ti,ab,kw OR (chang*):ti,ab,kw (Word variations have been searched)

#8 (diversi*):ti,ab,kw OR (heterogen*):ti,ab,kw OR (match*):ti,ab,kw OR (pair*):ti,ab,kw OR (different):ti,ab,kw OR (mutational burden):ti,ab,kw (Word variations have been searched)

#9 (congruen*):ti,ab,kw OR (incongruen*):ti,ab,kw OR (dyscongruen*):ti,ab,kw OR (discrepan*):ti,ab,kw OR (diverg*):ti,ab,kw OR (microsatellite instability):ti,ab,kw (Word variations have been searched)

#10 #3 OR #4 OR #5 OR #6

#11 #7 OR #8 OR #9

#12 (neoplas*):ti,ab,kw OR (tumor*):ti,ab,kw OR (cancer*):ti,ab,kw OR (carcinom*):ti,ab,kw OR (oncolog*):ti,ab,kw (Word variations have been searched)

#13 #2 OR #12

#14 (metasta*):ti,ab,kw OR (progressi*):ti,ab,kw (Word variations have been searched)

#15 #1 OR #14

#16 #15 AND #13 AND #10 AND #11

Web of Science

#1

(TS=((Neoplasm*) OR (Cancer*) OR (Carcinom*) OR (Malignant tumor*) OR (Oncolog*)))

#2

(TS=((Metasta*) OR (Progressi*) OR (Disseminat*) OR (Relaps*) OR (Recurr*)))

#3

(TS=((PD-1) OR (PD-L1) OR (PD-L2) OR (Programmed death ligand) OR (Tumor Infiltrating Lymphocyte*)))

#4

(TS=((Mutational Burden) OR (Microsatellite instability) OR (Alteration) OR (Altered) OR (Discordan*) OR (Concordan*) OR (Chang*) OR (Changing) OR (Congruen*) OR (Incongruen*) OR (Discrepan*) OR (Diversi*) OR (Variati*) OR (Difference*) OR (Differing) OR (Comparison) OR (Heterogen*) OR (mutational burden) OR (microsatellite instability)))

#5

(#4 AND #3 AND #2 AND #1)

AND LANGUAGE: (English) AND DOCUMENT TYPES: (Article)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	7



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	7
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7-8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7-8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-11
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8-11
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	8
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8-11
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	11
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	14
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	15

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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